5. $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
6. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
7. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
8. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
9. If $a+b+c=0$, then $a^3 + b^3 + c^3 = 3abc$

- **Pitfall:** Sign errors, incorrect order of operations (PEMDAS/BODMAS).
- **Technique:** Simplify expressions using identities, factorize, and solve equations systematically.

Alligation Mixture

Alligation is a rule that helps determine the ratio in which two or more ingredients of different prices must be mixed to produce a mixture of a desired price.

- **Formula:**

$$\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{Price of Dearer} - \text{Mean Price}}{\text{Mean Price} - \text{Price of Cheaper}}$$

- **Technique:** Use the "alligation cross" diagram to visualize and solve problems quickly.
- **Pitfall:** Incorrectly identifying the cheaper, dearer, or mean price.

Area

Area is the measure of the extent of a two-dimensional surface or shape.

- **Formulas:**
  1. **Square:** s^2 (where s is side)
  2. **Rectangle:** $l \times w$ (where l is length, w is width)
  3. **Circle:** πr^2 (where r is radius)
  4. **Triangle:** $\frac{1}{2} \times \text{base} \times \text{height}$ (or Heron's formula: $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$)
  5. **Parallelogram:** $\text{base} \times \text{height}$
  6. **Trapezoid:** $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
  7. **Rhombus:** $\frac{1}{2} \times d_1 \times d_2$ (where d_1, d_2 are diagonals)
- **Pitfall:** Using perimeter formulas instead of area, or incorrect units.
- **Technique:** Identify the shape, recall the correct formula, and ensure consistent units.

Arithmetic Series

An arithmetic series is the sum of terms in an arithmetic progression, where the difference between consecutive terms is constant (common difference d).

- **Formulas:**
  1. n-th term: $a_n = a_1 + (n-1)d$
  2. Sum of n terms: $S_n = \frac{n}{2}(a_1 + a_n)$ or $S_n = \frac{n}{2}(2a_1 + (n-1)d)$
- **Pitfall:** Off-by-one errors when calculating n, especially when given ranges.
- **Technique:** Clearly identify a_1, d, and n.

Average

The average (arithmetic mean) is the sum of a set of values divided by the number of values.

- **Formula:** $\text{Average} = \frac{\text{Sum of quantities}}{\text{Number of quantities}}$
- **Weighted Average:** $\text{Weighted Average} = \frac{\sum(w_i x_i)}{\sum w_i}$
- **Technique:** Use the formula directly. For weighted averages, ensure correct weights are applied.
- **Pitfall:** Confusing average with median or mode.

Bar Graph

A bar graph uses rectangular bars to represent data, with the length or height of the bars proportional to the values they represent.

- **Core Idea:** Visual representation of discrete data for comparison.
- **Technique:** Carefully read labels, scales, and legends. Compare bar heights/lengths to answer questions about quantities, differences, or ratios.
- **Pitfall:** Misinterpreting scales or units on the axes.

Bayes Theorem

Bayes' Theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

- **Formula:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **Key Terms:** $P(A|B)$ is posterior probability, $P(B|A)$ is likelihood, $P(A)$ is prior probability, $P(B)$ is evidence.
- **Pitfall:** Confusing conditional probabilities $P(A|B)$ and $P(B|A)$.
- **Technique:** Break down the problem into individual probabilities and substitute into the formula.

Calendar

Calendar problems involve determining the day of the week for a given date or finding specific dates based on calendar logic.

- **Core Idea:** Based on the concept of "odd days" (remainder when days are divided by 7).
- **Key Concepts:**
  1. Ordinary year = 365 days = 52 weeks + 1 odd day
  2. Leap year = 366 days = 52 weeks + 2 odd days
  3. Leap year rules: Divisible by 4, unless it's a century year not divisible by 400.
  4. Odd days for centuries: 100 years = 5 odd days, 200 years = 3, 300 years = 1, 400 years = 0.
- **Technique:** Calculate total odd days from a reference date (e.g., Jan 1, 0001) to the target date.
- **Pitfall:** Incorrectly identifying leap years or counting odd days.

Cartesian Coordinates

Cartesian coordinates define the position of a point in a plane using two perpendicular axes (x and y).

- **Key Formulas:**
  1. **Distance between** (x_1, y_1) **and** (x_2, y_2)**:** $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
  2. **Midpoint of a segment:** $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
  3. **Section Formula (dividing in ratio** $m : n$**):** $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
  4. **Area of triangle with vertices** $(x_1, y_1), (x_2, y_2), (x_3, y_3)$**:** $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- **Pitfall:** Sign errors in distance or section formula.
- **Technique:** Plot points if helpful, apply formulas carefully.

Circle

A circle is the set of all points in a plane that are equidistant from a central point.

- **Formulas:**
  1. **Circumference:** $C = 2\pi r = \pi d$
  2. **Area:** $A = \pi r^2$
  3. **Equation of a circle (center** (h, k)**, radius** r**):** $(x - h)^2 + (y - k)^2 = r^2$
  4. **Area of sector:** $\frac{\theta}{360^\circ} \times \pi r^2$ (where θ is central angle in degrees)
  5. **Length of arc:** $\frac{\theta}{360^\circ} \times 2\pi r$
- **Pitfall:** Confusing radius with diameter, or circumference with area.
- **Technique:** Identify given parameters and apply the correct formula.

Clock Time

Clock time problems involve calculating angles between clock hands, or determining exact times based on relative speeds of hands.

- **Key Rates:**
  1. Hour hand: 0.5° per minute
  2. Minute hand: 6° per minute
  3. Relative speed: 5.5° per minute (minute hand gains on hour hand)
- **Formula for angle between hands at H hours M minutes:** $|30H - \frac{11}{2}M|$
- **Technique:** Calculate the position of each hand relative to 12, then find the difference.
- **Pitfall:** Not considering the hour hand's movement within the hour.

Combinatory (Permutation and Combination)

Combinatory deals with counting arrangements (permutations) and selections (combinations) of objects.

- **Formulas:**
  1. **Permutation (arrangement of k items from n):** $P(n,k) = \frac{n!}{(n-k)!}$
  2. **Combination (selection of k items from n):** $C(n,k) = \frac{n!}{k!(n-k)!}$
  3. $n! = n \times (n-1) \times \cdots \times 1, 0! = 1$
- **Key Principle:** Permutation is about order, combination is not. "Arrangement" implies permutation, "selection" or "group" implies combination.
- **Pitfall:** Confusing permutation with combination.
- **Technique:** Determine if order matters. Use multiplication principle for sequential events, addition principle for mutually exclusive events.

Compound Interest

Compound interest is interest calculated on the initial principal, which also includes all of the accumulated interest from previous periods on a deposit or loan.

- **Formulas:**
  1. **Amount (compounded annually):** $A = P(1 + \frac{R}{100})^T$
  2. **Compound Interest (CI):** $CI = A - P$
  3. **Amount (compounded n times a year):** $A = P(1 + \frac{R}{100n})^{nT}$
  4. **Amount (compounded continuously):** $A = Pe^{RT/100}$
- **Pitfall:** Confusing simple interest with compound interest, or incorrect compounding frequency.
- **Technique:** Identify principal (P), rate (R), time (T), and compounding frequency.

Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

- **Formula:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (where $P(B) > 0$)
- **Key Idea:** The sample space is reduced to event B.
- **Pitfall:** Misidentifying the intersection event or the conditioning event.
- **Technique:** Clearly define events A and B, find their intersection, and the probability of B.

Cones

A cone is a three-dimensional geometric shape that tapers smoothly from a flat base (usually circular) to a point called the apex or vertex.

- **Formulas:**
  1. **Volume:** $V = \frac{1}{3}\pi r^2 h$
  2. **Curved Surface Area (CSA):** $CSA = \pi r l$ (where l is slant height, $l = \sqrt{r^2 + h^2}$)
  3. **Total Surface Area (TSA):** $TSA = \pi r(l + r)$
- **Pitfall:** Confusing height (h) with slant height (l).
- **Technique:** Use Pythagorean theorem to find l if r and h are given.

Contour Plots

Contour plots (or isoline maps) display three-dimensional data on a two-dimensional plane, where lines connect points of equal value.

- **Core Idea:** Visualize a third variable (e.g., elevation, temperature) over a surface.
- **Technique:** Interpret the density and values of contour lines. Closely spaced lines indicate a steep gradient, widely spaced lines indicate a gentle gradient.
- **Pitfall:** Misinterpreting the values represented by the lines or the gradient.

Cost Market Price

These problems involve concepts of cost price (CP), selling price (SP), market price (MP), profit, loss, and discount.

- **Formulas:**
  1. $\text{Profit} = \text{SP} - \text{CP}$ (if $\text{SP} > \text{CP}$)
  2. $\text{Loss} = \text{CP} - \text{SP}$ (if $\text{CP} > \text{SP}$)
  3. $\text{Profit/Loss} \backslash\% = \frac{\text{Profit/Loss}}{\text{CP}} \times 100$
  4. $\text{Discount} = \text{MP} - \text{SP}$
  5. $\text{Discount} \backslash\% = \frac{\text{Discount}}{\text{MP}} \times 100$
  6. $\text{SP} = \text{CP} \times (1 \pm \frac{\text{Profit/Loss} \backslash\%}{100})$
  7. $\text{SP} = \text{MP} \times (1 - \frac{\text{Discount} \backslash\%}{100})$
- **Pitfall:** Calculating profit/loss % on SP instead of CP, or discount % on CP instead of MP.
- **Technique:** Clearly identify CP, SP, MP, and the base for percentage calculations.

Counting

Counting principles involve determining the number of possible outcomes for various events.

- **Principles:**
  1. **Addition Principle:** If event A can occur in m ways and event B in n ways, and they are mutually exclusive, then A or B can occur in $m + n$ ways.
  2. **Multiplication Principle:** If event A can occur in m ways and event B in n ways, then A and B can occur in $m \times n$ ways.
- **Technique:** Break down complex counting problems into simpler, sequential or mutually exclusive steps.
- **Pitfall:** Overcounting or undercounting by misapplying the principles.

Cubes

A cube is a three-dimensional solid object bounded by six square faces, facets or sides, with three meeting at each vertex.

- **Formulas:**
  1. **Volume:** $V = a^3$ (where a is side length)
  2. **Surface Area:** $SA = 6a^2$
  3. **Diagonal of a face:** $a\sqrt{2}$
  4. **Main Diagonal of the cube:** $a\sqrt{3}$
- **Pitfall:** Confusing surface area with volume, or face diagonal with main diagonal.
- **Technique:** Visualize the cube and its dimensions.

Currency Notes

Problems involve calculating the total value of currency notes of different denominations or determining the number of notes of each denomination given a total amount.

- **Core Idea:** Form linear equations based on the value and count of notes.
- **Technique:** Assign variables to the number of notes of each denomination and set up equations.
- **Pitfall:** Calculation errors with large numbers or multiple denominations.

Curves

In quantitative aptitude, "curves" generally refer to the graphs of functions, requiring interpretation of their shape, slope, and points of interest.

- **Core Idea:** Visual representation of relationships between variables.
- **Technique:** Analyze the trend (increasing/decreasing), points of maxima/minima, and intersections.
- **Pitfall:** Misinterpreting the axes or the units.

Data Interpretation

Data interpretation involves extracting, analyzing, and drawing conclusions from various forms of data presentation like tables, charts (bar, pie, line, scatter, radar), and graphs.

- **Core Idea:** Critical analysis of presented data to answer specific questions.
- **Technique:** Read titles, labels, scales, and legends carefully. Perform calculations (percentages, ratios, averages) based on the data.
- **Pitfall:** Rushing calculations, misreading data points, or making assumptions not supported by the data.

Digital Image Processing

While a CS topic, in QA it might appear as basic numerical problems related to image size, resolution, or data storage.

- **Core Idea:** Image represented as a grid of pixels.
- **Key Concepts:**
  1. **Resolution:** Width $\times$ Height (in pixels)
  2. **Color Depth:** Bits per pixel (e.g., 8-bit for 256 colors, 24-bit for true color)
  3. **Image Size (uncompressed):** Resolution $\times$ Color Depth (in bits)
- **Technique:** Apply multiplication principles for calculating total bits/bytes.
- **Pitfall:** Unit conversions (bits to bytes, KB to MB).

Factors

Factors (or divisors) of a number are integers that divide the number evenly without leaving a remainder.

- **Core Idea:** Finding numbers that multiply to give the original number.
- **Properties:**
  1. For $N = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ (prime factorization):
  2. **Number of factors:** $(a_1 + 1)(a_2 + 1) \dots (a_k + 1)$
  3. **Sum of factors:** $(1 + p_1 + \dots + p_1^{a_1})(1 + p_2 + \dots + p_2^{a_2}) \dots$
- **Technique:** Prime factorization is key to finding number and sum of factors.
- **Pitfall:** Missing factors, especially 1 and the number itself.

Fractions

Fractions represent a part of a whole, expressed as a ratio of two integers (numerator/denominator).

- **Operations:** Addition, subtraction, multiplication, division.
- **Properties:** Equivalent fractions, proper/improper fractions, mixed numbers.
- **Technique:** Find common denominators for addition/subtraction. Invert and multiply for division. Simplify fractions.
- **Pitfall:** Errors in finding common denominators or simplifying.

Functions

A function is a relation between a set of inputs and a set of permissible outputs with the property that each input is related to exactly one output.

- **Core Idea:** Mapping inputs to outputs. $y = f(x)$.
- **Key Concepts:** Domain (set of inputs), Range (set of outputs), types (linear, quadratic, polynomial).
- **Technique:** Evaluate functions at given points, understand their graphs, and solve for unknowns.
- **Pitfall:** Incorrectly applying function rules or misinterpreting domain/range restrictions.

Geometry

Geometry deals with the properties and relations of points, lines, surfaces, solids, and higher dimensional analogs.

- **Core Idea:** Understanding shapes, angles, and spatial relationships.
- **Key Theorems:** Pythagorean theorem, properties of parallel lines and transversals, angle sum property of triangles, similar/congruent triangles.
- **Technique:** Draw diagrams, identify knowns and unknowns, apply relevant theorems and formulas.
- **Pitfall:** Assuming properties not explicitly stated or derivable.

Graph Coloring

Graph coloring assigns colors to elements of a graph subject to certain constraints. In QA, it's usually simplified to finding the minimum number of colors for a small graph.

- **Core Idea:** Assign colors to vertices such that no two adjacent vertices share the same color.
- **Technique:** Start coloring from a vertex with high degree; systematically assign colors, avoiding conflicts with neighbors.
- **Pitfall:** Missing a valid coloring or not finding the minimum.

Inequality

Inequalities are mathematical statements comparing two expressions using symbols like $<, >, \leq, \geq$.

- **Properties:**
  1. Adding/subtracting a number from both sides does not change the inequality direction.
  2. Multiplying/dividing by a positive number does not change the inequality direction.
  3. Multiplying/dividing by a negative number **reverses** the inequality direction.
- **Technique:** Solve like equations, but remember to flip the sign when multiplying/dividing by a negative number. For quadratic inequalities, use sign analysis of factors.
- **Pitfall:** Forgetting to reverse the inequality sign.

LCM HCF

LCM (Least Common Multiple) is the smallest positive integer divisible by each of a given set of integers. HCF (Highest Common Factor) or GCD (Greatest Common Divisor) is the largest positive integer that divides each of a given set of integers.

- **Formulas:**
  1. For two numbers a and b: $\text{LCM}(a, b) \times \text{HCF}(a, b) = a \times b$
  2. To find LCM/HCF: Use prime factorization method.
- **Technique:** Prime factorization is the most reliable method.
- **Pitfall:** Confusing LCM with HCF, especially in word problems.

Line Graph

A line graph displays information as a series of data points connected by straight line segments, showing trends over time or categories.

- **Core Idea:** Visualizing trends and changes over a continuous variable.
- **Technique:** Observe the slope of the lines to understand rates of change, identify peaks and troughs.
- **Pitfall:** Misinterpreting the scale or the meaning of the trend.

Lines

Lines in coordinate geometry are one-dimensional figures with properties like slope, intercepts, and equations.

- **Formulas:**
  1. **Slope** (m) **of a line through** (x_1, y_1) **and** (x_2, y_2)**:** $m = \frac{y_2 - y_1}{x_2 - x_1}$
  2. **Equation of a line (slope-intercept form):** $y = mx + c$ (where c is y-intercept)
  3. **Equation of a line (point-slope form):** $y - y_1 = m(x - x_1)$
  4. **Parallel lines:** $m_1 = m_2$
  5. **Perpendicular lines:** $m_1 m_2 = -1$
- **Pitfall:** Sign errors in slope calculation, confusing parallel with perpendicular conditions.

- **Technique:** Identify slope and a point, then use the appropriate equation form.

Logarithms

Logarithms are the inverse operation to exponentiation. The logarithm of a number x to the base b is the exponent to which b must be raised to produce x.

- **Definition:** $\log_b a = c \iff b^c = a$
- **Properties:**
  1. $\log_b(xy) = \log_b x + \log_b y$
  2. $\log_b(x/y) = \log_b x - \log_b y$
  3. $\log_b x^n = n \log_b x$
  4. $\log_b b = 1, \log_b 1 = 0$
  5. **Change of Base:** $\log_b a = \frac{\log_c a}{\log_c b}$
- **Pitfall:** Misapplying properties, especially for addition/subtraction.
- **Technique:** Convert between logarithmic and exponential forms, use properties to simplify.

Maps

Map problems involve interpreting scale, distances, and directions on maps.

- **Core Idea:** Scale represents the ratio of a distance on the map to the corresponding distance on the ground.
- **Formula:** $\text{Map Distance} = \text{Actual Distance} \times \text{Scale}$
- **Technique:** Pay close attention to the scale (e.g., 1:100000 or 1 cm = 10 km) and units.
- **Pitfall:** Incorrect unit conversions.

Maxima Minima

Finding the maximum or minimum value of a function or expression.

- **Core Idea:** Identifying the highest or lowest point a function can reach.
- **For Quadratic Function** $f(x) = ax^2 + bx + c$:
  1. Vertex at $x = -b/(2a)$
  2. If $a > 0$, minimum value at vertex.
  3. If $a < 0$, maximum value at vertex.
- **General Technique (Calculus-based, if applicable):** Find derivative $f'(x)$, set to zero to find critical points. Use second derivative test for max/min.
- **Pitfall:** Confusing local maxima/minima with global maxima/minima.

Mensuration

Mensuration is the branch of mathematics that deals with the measurement of geometric figures and their parameters like length, area, and volume. This topic consolidates formulas for various 2D and 3D shapes.

- **Core Idea:** Applying geometric formulas to calculate dimensions, areas, and volumes.
- **Technique:** Refer to specific topics like Area, Volume, Cones, Cubes, Circles, Triangles for detailed formulas.
- **Pitfall:** Incorrectly applying formulas or unit conversions.

Modular Arithmetic

Modular arithmetic is a system of arithmetic for integers, where numbers "wrap around" when reaching a certain value —the modulus.

- **Definition:** $a \equiv b \pmod{m}$ means a and b have the same remainder when divided by m, or m divides $(a-b)$.
- **Properties:**
  1. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.
  2. $(a \times b) \pmod{m} = ((a \pmod{m}) \times (b \pmod{m})) \pmod{m}$
- **Technique:** Use properties to simplify large numbers before finding remainders. Useful for unit digit problems.
- **Pitfall:** Incorrectly handling negative numbers in modulo operations.

Number Representation

Number representation deals with how numbers are written and interpreted, often involving different bases (e.g., decimal, binary).

- **Core Idea:** Converting numbers between different bases.
- **Technique:** For base-10 to base-B, repeatedly divide by B and collect remainders. For base-B to base-10, use positional notation (sum of digits $\times$ base to power of position).
- **Pitfall:** Errors in powers of the base or order of remainders.

Number Series

Number series problems involve identifying the pattern in a sequence of numbers and finding the next term or a missing term.

- **Types:** Arithmetic, geometric, difference series, squares/cubes, prime numbers, alternating patterns, Fibonacci-like.
- **Technique:** Look for common differences, common ratios, differences of differences, squares/cubes, or combinations of operations.
- **Pitfall:** Overlooking subtle patterns or assuming a simple pattern too quickly.

Number System

The number system classifies numbers (natural, whole, integers, rational, irrational, real, complex) and includes concepts like divisibility rules, prime/composite numbers.

- **Divisibility Rules:** (e.g., by 2, 3, 4, 5, 6, 8, 9, 10, 11).
- **Classification:**
  1. Natural Numbers: $\{1, 2, 3, \ldots\}$
  2. Whole Numbers: $\{0, 1, 2, 3, \ldots\}$
  3. Integers: $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$
  4. Rational Numbers: $\{p/q \mid p, q \text{ are integers}, q \neq 0\}$
  5. Irrational Numbers: Non-terminating, non-repeating decimals (e.g., $\sqrt{2}, \pi$)
  6. Real Numbers: Rational $+$ Irrational
- **Technique:** Apply divisibility rules, understand properties of number types.
- **Pitfall:** Confusing different sets of numbers (e.g., integers vs. whole numbers).

Number Theory

Number theory is a branch of pure mathematics devoted primarily to the study of integers and integer-valued functions, including primes, divisibility, and modular arithmetic.

- **Core Idea:** Properties of integers. (Overlaps with Factors, Prime Numbers, Modular Arithmetic).
- **Key Concepts:** Prime numbers, composite numbers, factors, multiples, HCF, LCM, divisibility.
- **Technique:** Use prime factorization, apply divisibility rules, understand modular arithmetic.
- **Pitfall:** Misapplying theorems or properties.

Numerical Computation

Numerical computation involves performing calculations accurately and efficiently, often with approximations, rounding, and significant figures.

- **Core Idea:** Precision and accuracy in calculations.
- **Technique:** Estimate answers, use approximations (e.g., $\pi \approx 22/7$ or 3.14), understand rules for significant figures and rounding.
- **Pitfall:** Rounding too early, leading to accumulated errors.

Percentage

Percentage is a way of expressing a number or ratio as a fraction of 100. It is often denoted using the percent sign, "%".

- **Formulas:**
  1. $\text{Value} = \text{Base} \times \frac{\text{Percentage}}{100}$
  2. $\text{Percentage Change} = \frac{\text{Change}}{\text{Original Value}} \times 100$

3. Successive Percentage Change: If an item changes by $x\%$ then by $y\%$, net change is $(x+y+\frac{xy}{100})\%$.
- **Pitfall:** Calculating percentage change based on the wrong base value.
- **Technique:** Convert percentages to decimals or fractions for calculations.

Permutation and Combination

Refer to "Combinatory" section for details.

Pie Chart

A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportion.

- **Core Idea:** Visualizing parts of a whole as proportions.
- **Properties:** The sum of all percentages in a pie chart must be 100%, and the sum of all central angles must be $360°$.
- **Technique:** Relate percentages to degrees ($1\% = 3.6°$). Calculate proportions and compare sectors.
- **Pitfall:** Misinterpreting the size of sectors or the total quantity.

Polynomials

A polynomial is an expression consisting of variables and coefficients, that involves only the operations of addition, subtraction, multiplication, and non-negative integer exponents of variables.

- **Key Concepts:** Degree, roots (zeros), factors.
- **Theorems:**
  1. **Remainder Theorem:** If a polynomial $P(x)$ is divided by $(x-a)$, the remainder is $P(a)$.
  2. **Factor Theorem:** $(x-a)$ is a factor of $P(x)$ if and only if $P(a)=0$.
- **Technique:** Factorization, synthetic division, applying remainder/factor theorems.
- **Pitfall:** Errors in algebraic manipulation or sign errors.

Powers

Powers (or exponents) indicate the number of times a base number is multiplied by itself.

- **Rules of Exponents:**
  1. $x^a \cdot x^b = x^{a+b}$
  2. $x^a / x^b = x^{a-b}$
  3. $(x^a)^b = x^{ab}$
  4. $(xy)^a = x^a y^a$
  5. $(x/y)^a = x^a / y^a$
  6. $x^0 = 1$ (for $x \neq 0$)
  7. $x^{-a} = 1/x^a$
  8. $x^{1/n} = \sqrt[n]{x}$
- **Pitfall:** Incorrectly applying rules, especially with negative exponents or fractions.
- **Technique:** Simplify expressions using exponent rules, convert to common base if possible.

Prime Numbers

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

- **Properties:**
  1. 2 is the only even prime number.
  2. Every integer greater than 1 is either prime or can be uniquely expressed as a product of primes (Fundamental Theorem of Arithmetic).
- **Technique:** To check if N is prime, test divisibility by primes up to $\sqrt{N}$.
- **Pitfall:** Confusing 1 as a prime number (it's not).

Probability

Probability is the measure of the likelihood that an event will occur.

- **Formula:** $P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$
- **Properties:**
  1. $0 \leq P(E) \leq 1$
  2. $P(\text{not } E) = 1 - P(E)$
  3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  4. For independent events: $P(A \cap B) = P(A)P(B)$
- **Technique:** Clearly define the sample space and the event. Use counting techniques (P&C) to find favorable and total outcomes.
- **Pitfall:** Incorrectly identifying sample space or favorable outcomes, or assuming independence when events are dependent.

Probability Density Function

For continuous random variables, the Probability Density Function (PDF) describes the relative likelihood for the random variable to take on a given value. The probability of the variable falling within a range is given by the integral of the PDF over that range.

- **Core Idea:** For continuous variables, $P(a \leq X \leq b) = \int_a^b f(x)dx$.
- **Property:** $\int_{-\infty}^{\infty} f(x)dx = 1$
- **Technique:** Understand that for continuous variables, $P(X = x) = 0$. Focus on areas under the curve.
- **Pitfall:** Treating PDF values directly as probabilities (they are not).

Profit Loss

Refer to "Cost Market Price" section for details.

Quadratic Equations

A quadratic equation is a polynomial equation of the second degree, typically written as $ax^2 + bx + c = 0$.

- **Formulas:**
  1. **Quadratic Formula (roots):** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
  2. **Discriminant:** $D = b^2 - 4ac$
     - If $D > 0$, two distinct real roots.
     - If $D = 0$, two equal real roots.
     - If $D < 0$, no real roots (complex roots).
  3. **Sum of roots:** $\alpha + \beta = -b/a$
  4. **Product of roots:** $\alpha\beta = c/a$
- **Technique:** Factorization, completing the square, or using the quadratic formula.
- **Pitfall:** Sign errors in the quadratic formula or discriminant.

Radar Chart

A radar chart (or spider chart) displays multivariate data in the form of a two-dimensional chart of three or more quantitative variables represented on axes starting from the same point.

- **Core Idea:** Comparing multiple attributes of one or more items.
- **Technique:** Compare the area covered by different polygons or the values along each axis.
- **Pitfall:** Overcrowding with too many variables or items, making comparisons difficult.

Ratio Proportion

Ratio is a comparison of two quantities by division. Proportion states that two ratios are equal.

- **Formulas:**
  1. **Ratio:** $a : b = a/b$
  2. **Proportion:** $a : b :: c : d \implies a/b = c/d \implies ad = bc$ (product of extremes = product of means)
  3. **Direct Proportion:** $y = kx$
  4. **Inverse Proportion:** $y = k/x$

- **Technique:** Simplify ratios, use cross-multiplication for proportions, apply unitary method for direct/inverse proportion problems.
- **Pitfall:** Incorrectly setting up ratios or proportions, especially in word problems.

Scatter Plot

A scatter plot uses Cartesian coordinates to display values for two variables for a set of data. The data is displayed as a collection of points, each having the value of one variable determining the position on the horizontal axis and the value of the other variable determining the position on the vertical axis.

- **Core Idea:** Visualizing the relationship (correlation) between two variables.
- **Technique:** Observe the trend of points (positive, negative, or no correlation), strength of correlation (tightness of points).
- **Pitfall:** Inferring causation from correlation.

Seating Arrangement

Seating arrangement problems involve arranging people or objects in a specific order (linear, circular, etc.) based on given conditions.

- **Core Idea:** Logical deduction and application of permutation principles.
- **Technique:** Draw diagrams, start with fixed positions or most restrictive conditions, use elimination. For circular arrangements, fix one person's position first.
- **Pitfall:** Missing a condition, or misinterpreting "left/right" in circular arrangements.

Sequence Series

A sequence is an ordered list of numbers. A series is the sum of the terms of a sequence. (Refer to Arithmetic Series for specific formulas).

- **Core Idea:** Identifying patterns and calculating sums.
- **Types:** Arithmetic, Geometric, Harmonic, special series (squares, cubes).
- **Geometric Series:**
  1. n-th term: $a_n = ar^{n-1}$
  2. Sum of n terms: $S_n = \frac{a(r^n-1)}{r-1}$ (for $r \neq 1$)
  3. Sum to infinity: $S_\infty = \frac{a}{1-r}$ (for $|r| < 1$)
- **Technique:** Determine the type of sequence, find the common difference/ratio, and apply the correct formula.
- **Pitfall:** Confusing arithmetic and geometric series formulas.

Set Theory

Set theory deals with collections of objects (sets) and their properties and operations.

- **Key Concepts:** Union ($\cup$), Intersection ($\cap$), Complement (A' or A^c), Difference ($A - B$), Subset ($\subseteq$), Cardinality ($|A|$).
- **Formulas:**
  1. $|A \cup B| = |A| + |B| - |A \cap B|$
  2. $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$
- **Technique:** Use Venn diagrams to visualize and solve problems involving multiple sets.
- **Pitfall:** Double-counting or missing elements in set operations.

Shortest Path

While a CS algorithm topic, in QA, it might involve finding the shortest distance between two points on a simple grid or a small, explicitly given graph.

- **Core Idea:** Finding the minimum distance or cost to travel between two nodes.
- **Technique:** For simple graphs, inspect paths and sum edge weights. For grids, use Manhattan distance or Euclidean distance depending on movement rules.
- **Pitfall:** Missing a shorter path or incorrectly calculating path length.

Speed Time Distance

These problems relate speed, time, and distance, often involving relative motion, trains, or boats.

- **Formulas:**
  1. $\text{Distance} = \text{Speed} \times \text{Time}$
  2. $\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$
  3. **Relative Speed (same direction):** $|S_1 - S_2|$
  4. **Relative Speed (opposite direction):** $S_1 + S_2$
  5. Conversion: $1 \text{ km/hr} = 5/18 \text{ m/s}$
- **Technique:** Ensure consistent units. For relative speed, determine if objects are moving towards or away from each other.
- **Pitfall:** Unit conversion errors, or misapplying relative speed concepts.

Squares

A square is a regular quadrilateral, which means it has four equal sides and four equal angles ($90°$ each).

- **Formulas:**
  1. **Area:** s^2
  2. **Perimeter:** $4s$
  3. **Diagonal:** $s\sqrt{2}$
- **Properties:** All sides equal, all angles $90°$, diagonals bisect each other at $90°$.
- **Technique:** Apply formulas directly. Recognize perfect squares.
- **Pitfall:** Confusing square properties with other quadrilaterals.

Statistics

Statistics involves the collection, analysis, interpretation, presentation, and organization of data.

- **Key Measures:**
  1. **Mean:** Average of all values.
  2. **Median:** Middle value when data is ordered.
  3. **Mode:** Most frequent value.
  4. **Range:** Max value - Min value.
  5. **Standard Deviation:** Measure of data dispersion (basic understanding).
- **Technique:** Calculate measures of central tendency and dispersion. Interpret basic statistical graphs.
- **Pitfall:** Confusing mean, median, and mode.

System of Equations

A system of equations is a set of two or more equations containing common variables, which are to be solved simultaneously.

- **Core Idea:** Finding values for variables that satisfy all equations simultaneously.
- **Techniques:**
  1. **Substitution:** Solve one equation for a variable, substitute into others.
  2. **Elimination:** Add/subtract equations to eliminate a variable.
  3. **Matrix methods:** For larger systems (less common in basic QA).
- **Pitfall:** Algebraic errors during substitution or elimination.

Tables / Tabular Data

Tables organize data into rows and columns, providing a structured way to present information.

- **Core Idea:** Extracting specific data points and performing calculations based on row/column totals or individual entries.
- **Technique:** Carefully read row and column headers, identify the required data, and perform calculations (sum, average, percentage, ratio).
- **Pitfall:** Misreading data from the wrong row/column, or incorrect calculations.

Triangles

A triangle is a polygon with three edges and three vertices. It is one of the basic shapes in geometry.

- **Properties:**
  1. Sum of angles = 180°.
  2. Triangle Inequality: Sum of any two sides is greater than the third side.
  3. Types: Equilateral, Isosceles, Scalene, Right-angled, Acute, Obtuse.
- **Formulas:**
  1. **Area:** $\frac{1}{2} \times \text{base} \times \text{height}$ (or Heron's formula)
  2. **Pythagorean Theorem (for right triangle):** $a^2 + b^2 = c^2$
- **Technique:** Draw diagrams, identify triangle type, apply relevant theorems (e.g., Pythagorean, similar triangles).
- **Pitfall:** Assuming a triangle is right-angled or isosceles without sufficient information.

Trigonometry

Trigonometry deals with the relationships between the sides and angles of triangles, particularly right-angled triangles.

- **Ratios (SOH CAH TOA):**
  1. $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$
  2. $\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$
  3. $\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{\sin\theta}{\cos\theta}$
- **Identities:**
  1. $\sin^2\theta + \cos^2\theta = 1$
  2. $\sec^2\theta - \tan^2\theta = 1$
  3. $\csc^2\theta - \cot^2\theta = 1$
- **Standard Angles:** Memorize values for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$.
- **Technique:** Draw right triangles, label sides, apply appropriate ratios.
- **Pitfall:** Confusing ratios, or incorrect use of identities.

Unit Digit

Finding the unit digit of large powers or complex expressions.

- **Core Idea:** The unit digit of powers of a number follows a cycle.
- **Technique:** Identify the cyclicity of the unit digit of the base number (e.g., 2: 2,4,8,6; 3: 3,9,7,1; 4: 4,6; 5: 5; 6: 6; 7: 7,9,3,1; 8: 8,4,2,6; 9: 9,1). Divide the exponent by the cycle length and use the remainder.
- **Pitfall:** Errors in determining the cycle or using the remainder.

Venn Diagram

Venn diagrams are graphical representations used to show relationships between sets. (Refer to Set Theory for formulas).

- **Core Idea:** Visualizing set operations and relationships.
- **Technique:** Draw overlapping circles (or other shapes) to represent sets. Fill in the numbers in each region based on the given information, starting from the innermost intersection.
- **Pitfall:** Incorrectly placing numbers in regions or misinterpreting the meaning of overlapping areas.

Volume

Volume is the amount of three-dimensional space occupied by an object or substance.

- **Formulas:**
  1. **Cube:** a^3
  2. **Cuboid:** $l \times w \times h$
  3. **Cylinder:** $\pi r^2 h$
  4. **Cone:** $\frac{1}{3}\pi r^2 h$
  5. **Sphere:** $\frac{4}{3}\pi r^3$

6. **Hemisphere:** $\frac{2}{3}\pi r^3$

- **Technique:** Identify the 3D shape, recall the correct formula, ensure consistent units.
- **Pitfall:** Confusing volume with surface area, or using incorrect dimensions.

Work Time

Work-time problems involve calculating the time taken to complete a task by individuals or groups, often working at different rates.

- **Core Idea:** Work = Rate × Time. Rate is typically "work per unit time".
- **Formulas:**
  1. If A takes x days, A's 1-day work = $1/x$.
  2. If A and B work together, their combined 1-day work = $1/x + 1/y$.
  3. $\text{Efficiency} \propto \frac{1}{\text{Time}}$
  4. $\text{Men} \times \text{Days} \times \text{Hours}/\text{Work} = \text{Constant}$ (for multiple workers)
- **Technique:** Convert all work rates to a common unit (e.g., work per day). Use LCM method for efficiency.
- **Pitfall:** Incorrectly adding/subtracting rates, or misinterpreting "together" vs. "alone" work.

Quick Formula Reference

- **Absolute Value:** $|x| = a \implies x = \pm a$; \(|x|a \implies x<-a \text{ or } x>a\)
- **Algebraic Identities:** $(a+b)^2 = a^2 + 2ab + b^2$, $a^2 - b^2 = (a-b)(a+b)$, $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- **Alligation:** $\frac{Q_C}{Q_D} = \frac{P_D - P_M}{P_M - P_C}$
- **Area:** Square s^2, Rectangle lw, Circle πr^2, Triangle $\frac{1}{2}bh$
- **Arithmetic Series:** $a_n = a_1 + (n-1)d$, $S_n = \frac{n}{2}(a_1 + a_n)$
- **Average:** $\frac{\sum x}{n}$
- **Bayes Theorem:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **Cartesian Coordinates:** Distance $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, Midpoint $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
- **Circle:** Circumference $2\pi r$, Area πr^2
- **Clock Angle:** $|30H - \frac{11}{2}M|$
- **Combinations:** $C(n,k) = \frac{n!}{k!(n-k)!}$
- **Permutations:** $P(n,k) = \frac{n!}{(n-k)!}$
- **Compound Interest:** $A = P(1 + R/100)^T$
- **Conditional Probability:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- **Cone:** Volume $\frac{1}{3}\pi r^2 h$, CSA $\pi r l$, TSA $\pi r(l+r)$
- **Cost/Profit/Loss:** $\text{Profit/Loss }\backslash\% = \frac{\text{Profit/Loss}}{\text{CP}} \times 100$, $\text{Discount }\backslash\% = \frac{\text{Discount}}{\text{MP}} \times 100$
- **Cube:** Volume a^3, Surface Area $6a^2$
- **Factors:** For $N = p_1^{a_1} \dots p_k^{a_k}$, No. of factors $(a_1+1)\dots(a_k+1)$
- **Geometric Series:** $a_n = ar^{n-1}$, $S_n = \frac{a(r^n-1)}{r-1}$
- **LCM HCF:** $\text{LCM}(a,b) \times \text{HCF}(a,b) = a \times b$
- **Lines:** Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$, $y = mx + c$, Parallel $m_1 = m_2$, Perpendicular $m_1 m_2 = -1$
- **Logarithms:** $\log(xy) = \log x + \log y$, $\log(x/y) = \log x - \log y$, $\log x^n = n \log x$
- **Percentage:** $\text{Value} = \text{Base} \times \frac{\text{Percentage}}{100}$
- **Probability:** $P(E) = \frac{\text{Favorable}}{\text{Total}}$
- **Quadratic Equation:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, Sum of roots $-b/a$, Product of roots c/a
- **Ratio Proportion:** $a : b :: c : d \implies ad = bc$
- **Set Theory:** $|A \cup B| = |A| + |B| - |A \cap B|$
- **Speed Time Distance:** $D = S \times T$, Relative Speed (same) $|S_1 - S_2|$, (opposite) $S_1 + S_2$
- **Trigonometry:** $\sin\theta = O/H$, $\cos\theta = A/H$, $\tan\theta = O/A$, $\sin^2\theta + \cos^2\theta = 1$
- **Volume:** Cube a^3, Cuboid lwh, Cylinder $\pi r^2 h$, Cone $\frac{1}{3}\pi r^2 h$, Sphere $\frac{4}{3}\pi r^3$
- **Work Time:** Work = Rate × Time

Important Tips for GATE

1. **Master Fundamentals:** Ensure a strong grasp of basic arithmetic, algebra, geometry, and data interpretation. Many complex problems are built upon these foundational concepts.
2. **Memorize Formulas:** Quantitative Aptitude is heavily formula-driven. Create flashcards or a dedicated notebook for all key formulas and identities. Regular revision is crucial.
3. **Practice Data Interpretation:** DI questions are common and can be time-consuming. Practice reading various charts (bar, pie, line, scatter, radar) and tables accurately and quickly. Pay attention to scales and units.
4. **Time Management:** Aptitude questions often have quick solutions if the right approach is identified. Don't get stuck on one question; if it's taking too long, mark it for review and move on.
5. **Unit Consistency:** Always check and convert units to be consistent within a problem (e.g., km/hr to m/s, cm to meters) to avoid common calculation errors.
6. **Approximation Skills:** For questions with close options or complex calculations, develop the ability to estimate and approximate to quickly narrow down choices.
7. **Identify Question Type:** Quickly categorize the problem (e.g., P&C, Profit/Loss, Speed-Time) to recall the relevant formulas and problem-solving strategies.
8. **Review Common Pitfalls:** Be aware of typical mistakes like sign errors in algebra, confusing permutations with combinations, or misinterpreting relative speed. Consciously check for these during practice.

9.1 No Classified Topic (1)

9.1.1 GATE CSE 2026 | Set 2 | GA | Question: 4

The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. _______

A. 0
B. 5
C. 10
D. 20

gatecse-2026-set2 quantitative-aptitude one-mark

Answer key

9.2 Absolute Value (5)

✍ **Practice Test:** Test 1 (10Q)

9.2.1 Absolute Value: GATE CSE 2014 | Set 2 | GA | Question: 8

If x is real and $|x^2-2x+3|=11$, then possible values of $|-x^3+x^2-x|$ include

A. $2,4$
B. $2,14$
C. $4,52$
D. $14,52$

gatecse-2014-set2 quantitative-aptitude normal absolute-value

Answer key

9.2.2 Absolute Value: GATE CSE 2017 | Set 1 | GA | Question: 8

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to:

A. The maximum of x and y
B. The minimum of x and y
C. 1
D. None of the above

gatecse-2017-set1 general-aptitude quantitative-aptitude maxima-minima absolute-value easy

Answer key

9.2.3 Absolute Value: GATE2011 AG: GA-7

Given that $f(y)=\frac{|y|}{y}$, and q is non-zero real number, the value of $|f(q)-f(-q)|$ is

A. 0
B. -1
C. 1
D. 2

general-aptitude quantitative-aptitude gate2011-ag absolute-value

Answer key

9.2.4 Absolute Value: GATE2013 AE: GA-8

If $|-2X+9|=3$ then the possible value of $|-X|-X^2$ would be:

A. 30 B. -30 C. -42 D. 42

gate2013-ae quantitative-aptitude absolute-value

Answer key

9.2.5 Absolute Value: GATE2013 CE: GA-7

If $|4X-7|=5$ then the values of $2\,|X|-|-X|$ is:

A. $2, \left(\frac{1}{3}\right)$ B. $\left(\frac{1}{2}\right), 3$ C. $\left(\frac{3}{2}\right), 9$ D. $\left(\frac{2}{3}\right), 9$

gate2013-ce quantitative-aptitude absolute-value

Answer key

9.3 Algebra (1)

Practice Tests: Test 1 (15Q) Test 2 (12Q)

9.3.1 Algebra: GATE2011 MN: GA-61

If $\frac{(2y+1)}{(y+2)} < 1$, then which of the following alternatives gives the CORRECT range of y ?

A. $-2 < y < 2$
B. $-2 < y < 1$
C. $-3 < y < 1$
D. $-4 < y < 1$

quantitative-aptitude gate2011-mn algebra

Answer key

9.4 Alligation Mixture (2)

Practice Test: Test 1 (8Q)

9.4.1 Alligation Mixture: GATE CSE 2011 | Question: 65

A container originally contains 10 litres of pure spirit. From this container, 1 litre of spirit replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

A. 7.58 litres
B. 7.84 litres
C. 7 litres
D. 7.29 litres

gatecse-2011 quantitative-aptitude normal alligation-mixture

Answer key

9.4.2 Alligation Mixture: GATE2010 TF: GA-9

A tank has 100 liters of water. At the end of every hour, the following two operations are performed in sequence: i) water equal to $m\%$ of the current contents of the tank is added to the tank, ii) water equal to $n\%$ of the current contents of the tank is removed from the tank. At the end of 5 hours, the tank contains exactly 100 liters of water. The relation between m and n is :

A. $m = n$
B. $m > n$
C. $m < n$
D. None of the previous

general-aptitude quantitative-aptitude gate2010-tf alligation-mixture

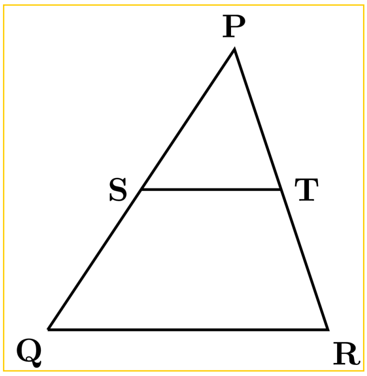
Answer key

9.5 Area (1)

9.5.1 Area: GATE CSE 2025 | Set 1 | GA | Question: 7

In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

Note: The figure shown is representative.



A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{2}{5}$ D. $\frac{1}{2}$

gatecse2025-set1 quantitative-aptitude geometry area two-marks

Answer key

9.6 Arithmetic Series (3)

Practice Test: Test 1 (10Q)

9.6.1 Arithmetic Series: GATE CSE 2013 | Question: 58

What will be the maximum sum of $44, 42, 40, \ldots$?

A. 502 B. 504 C. 506 D. 500

gatecse-2013 quantitative-aptitude easy arithmetic-series

Answer key

9.6.2 Arithmetic Series: GATE CSE 2015 | Set 2 | GA | Question: 6

If the list of letters $P,\ R,\ S,\ T,\ U$ is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. $2P, 2R, 2S, 2T, 2U$
II. $P-3, R-3, S-3, T-3, U-3$
III. P^2, R^2, S^2, T^2, U^2

A. I only B. I and II C. II and III D. I and III

gatecse-2015-set2 quantitative-aptitude normal arithmetic-series

Answer key

9.6.3 Arithmetic Series: GATE2011 AG: GA-6

The sum of n terms of the series $4 + 44 + 444 + \ldots\ldots$ is

A. $\frac{4}{81}\left[10^{n+1} - 9n - 1\right]$

B. $\frac{4}{81}\left[10^{n-1} - 9n - 1\right]$

C. $\frac{4}{81}\left[10^{n+1}-9n-10\right]$

D. $\frac{4}{81}\left[10^{n}-9n-10\right]$

general-aptitude quantitative-aptitude gate2011-ag arithmetic-series

Answer key

9.7 Average (1)

Practice Test: Test 1 (13Q)

9.7.1 Average: GATE CSE 2025 | Set 1 | GA | Question: 3

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

A. 25 B. 28 C. 30 D. 32

gatecse2025-set1 quantitative-aptitude average one-mark

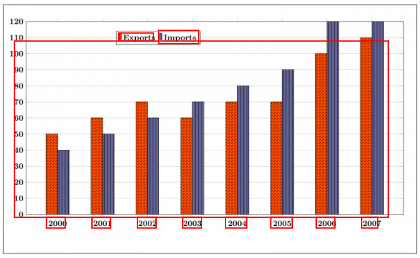
Answer key

9.8 Bar Graph (4)

Practice Tests: Test 1 (15Q) Test 2 (4Q)

9.8.1 Bar Graph: GATE CSE 2015 | Set 3 | GA | Question: 10

The exports and imports (in crores of $Rs.$) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?

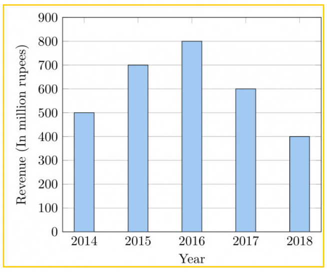


gatecse-2015-set3 quantitative-aptitude data-interpretation normal numerical-answers bar-graph

Answer key

9.8.2 Bar Graph: GATE CSE 2020 | GA | Question: 10

The total revenue of a company during $2014-2018$ is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during $2014-2018$ is ____________.

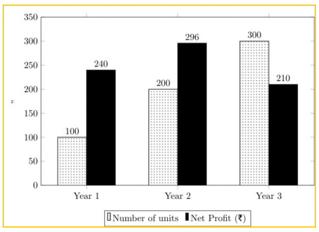


A. 16.67% profit B. 16.67% loss
C. 20% profit D. 20% loss

gatecse-2020 quantitative-aptitude data-interpretation bar-graph two-marks

Answer key

9.8.3 Bar Graph: GATE CSE 2021 | Set 2 | GA | Question: 9



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹ 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13%, and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is __________.

A. $4:3$ B. $1:1$ C. $3:4$ D. $1:2$

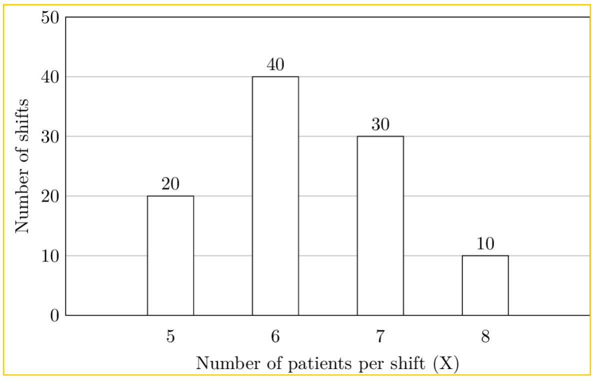
gatecse-2021-set2 quantitative-aptitude data-interpretation bar-graph two-marks

Answer key

9.8.4 Bar Graph: GATE DA 2025 | GA | Question: 10

The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is ₹ $1000(X-0.2)$, what is the average amount (in ₹) she has earned per shift in the past 100 shifts?

Note: The figure shown is representative.



A. $6,100$ B. $6,300$ C. $6,000$ D. $6,500$

gateda-2025 quantitative-aptitude bar-graph data-interpretation two-marks

Answer key

9.9 Bayes Theorem (1)

9.9.1 Bayes Theorem: GATE CSE 2012 | Question: 63

An automobile plant contracted to buy shock absorbers from two suppliers X and Y . X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of $X's$ shock absorbers, 96% are reliable. Of $Y's$ shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 B. 0.334 C. 0.667 D. 0.720

gatecse-2012 quantitative-aptitude bayes-theorem probability normal conditional-probability

Answer key

9.10 Cartesian Coordinates (4)

Practice Test: Test 1 (10Q)

9.10.1 Cartesian Coordinates: GATE CSE 2020 | GA | Question: 9

Two straight lines are drawn perpendicular to each other in $X-Y$ plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha+\beta$ is ________.

A. 60° B. 90° C. 120° D. 180°

gatecse-2020 quantitative-aptitude geometry cartesian-coordinates two-marks

Answer key

9.10.2 Cartesian Coordinates: GATE CSE 2022 | GA | Question: 2

A function $y(x)$ is defined in the interval $[0,1]$ on the $x-$axis as

$$y(x)=\begin{cases} 2 & \text{if} \quad 0 \leq x < \frac{1}{3} \\ 3 & \text{if} \quad \frac{1}{3} \leq x < \frac{3}{4} \\ 1 & \text{if} \quad \frac{3}{4} \leq x \leq 1 \end{cases}$$

Which one of the following is the area under the curve for the interval $[0,1]$ on the $x-$axis?

A. $\frac{5}{6}$ B. $\frac{6}{5}$ C. $\frac{13}{6}$ D. $\frac{6}{13}$

gatecse-2022 quantitative-aptitude cartesian-coordinates area one-mark

Answer key

9.10.3 Cartesian Coordinates: GATE2012 AE: GA-9

Two points $(4,p)$ and $(0,q)$ lie on a straight line having a slope of $3/4$. The value of $(p-q)$ is

A. -3 B. 0 C. 3 D. 4

gate2012-ae quantitative-aptitude cartesian-coordinates geometry

Answer key

9.10.4 Cartesian Coordinates: GATE2014 AE: GA-4

If $y=5x^2+3$, then the tangent at $x=0$, $y=3$

A. passes through $x=0, y=0$
B. has a slope of $+1$
C. is parallel to the x-axis
D. has a slope of -1

gate2014-ae quantitative-aptitude geometry cartesian-coordinates

Answer key

9.11 Circle (3)

Practice Test: Test 1 (10Q)

9.11.1 Circle: GATE CSE 2018 | GA | Question: 3

The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

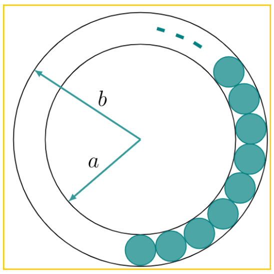
A. πd B. πd^2 C. $\frac{1}{4}\pi d^2$ D. $\frac{1}{2}\pi d$

gatecse-2018 quantitative-aptitude geometry circle normal one-mark

Answer key

9.11.2 Circle: GATE CSE 2020 | GA | Question: 8

The figure below shows an annular ring with outer and inner as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is ______.



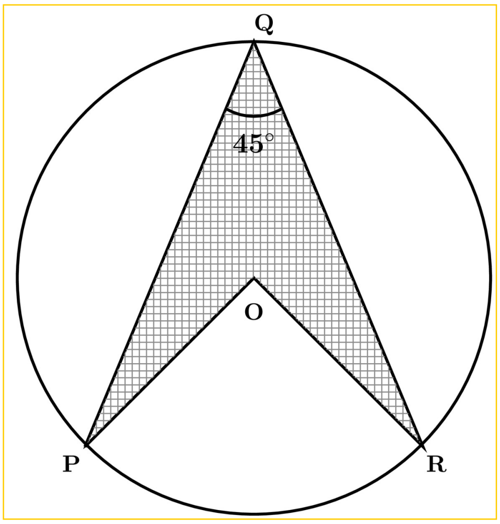
A. $\pi[(b^2 - a^2) - \frac{n}{4}(b-a)^2]$
B. $\pi[(b^2 - a^2) - n(b-a)^2]$
C. $\pi[(b^2 - a^2) + \frac{n}{4}(b-a)^2]$
D. $\pi[(b^2 - a^2) + n(b-a)^2]$

gatecse-2020 quantitative-aptitude geometry circle area two-marks

Answer key

9.11.3 Circle: GATE DA 2026 | GA | Question: 10

In the given figure, P, Q, and R are three points on a circle of radius 10 cm with O as its center, $PQ = RQ$, and $\angle PQR = 45°$. The figure is representative. The area of the shaded region $PQRO$ is _____ cm^2.



A. 50
B. $25\sqrt{2}$
C. $50\sqrt{2}$
D. 100

gateda-2026 quantitative-aptitude geometry circle area two-marks

Answer key

9.12 Clock Time (1)

✍ **Practice Test:** Test 1 (11Q)

9.12.1 Clock Time: GATE CSE 2014 | Set 2 | GA | Question: 10

At what time between 6 a. m. and 7 a. m. will the minute hand and hour hand of a clock make an angle closest to $60°$?

A. $6:22$ a.m. B. $6:27$ a.m. C. $6:38$ a.m. D. $6:45$ a.m.

gatecse-2014-set2 quantitative-aptitude normal clock-time

Answer key

9.13 Combinatory (5)

✍ **Practice Test:** Test 1 (13Q)

9.13.1 Combinatory: GATE CSE 2010 | Question: 65

Given digits $2, 2, 3, 3, 3, 4, 4, 4, 4$ how many distinct 4 digit numbers greater than 3000 can be formed?

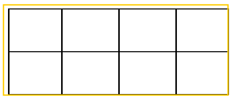
A. 50 B. 51 C. 52 D. 54

gatecse-2010 quantitative-aptitude combinatory normal

Answer key

9.13.2 Combinatory: GATE CSE 2016 | Set 2 | GA | Question: 09

In a 2×4 rectangle grid shown below, each cell is rectangle. How many rectangles can be observed in the grid?



A. 21 B. 27 C. 30 D. 36

gatecse-2016-set2 quantitative-aptitude normal combinatory

Answer key

9.13.3 Combinatory: GATE CSE 2017 | Set 1 | GA | Question: 9

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

A. 21 B. 18 C. 16 D. 14

gatecse-2017-set1 combinatory quantitative-aptitude

Answer key

9.13.4 Combinatory: GATE2011 MN: GA-59

In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

A. 4 B. 12 C. 64 D. 81

quantitative-aptitude gate2011-mn combinatory

Answer key

9.13.5 Combinatory: GATE2012 AR: GA-5

Ten teams participate in a tournament. Every team plays each of the other teams twice. The total number of matches to be played is

A. 20 B. 45 C. 60 D. 90

gate2012-ar quantitative-aptitude combinatory

Answer key

9.14 Compound Interest (3)

✍ **Practice Test:** Test 1 (5Q)

9.14.1 Compound Interest: GATE DS&AI 2024 | GA | Question: 8

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund A	Mutual fund B	Mutual fund C
Person1	10,000	20,000	20,000
Person2	20,000	15,000	15,000

At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

A. 7.5% B. 10% C. 15% D. 20%

gate-ds-ai-2024 quantitative-aptitude compound-interest two-marks

Answer key

9.14.2 Compound Interest: GATE2010 MN: GA-5

A person invest Rs.1000 at 10% annual compound interest for 2 years. At the end of two years, the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is :

A. 1776 B. 1760 C. 1920 D. 1936

general-aptitude quantitative-aptitude gate2010-mn compound-interest

Answer key

9.14.3 Compound Interest: GATE2014 AG: GA-5

The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

A. $3-4$ years B. $4-5$ years C. $5-6$ years D. $6-7$ years

gate2014-ag quantitative-aptitude compound-interest normal

Answer key

9.15 Conditional Probability (2)

✍ **Practice Test:** Test 1 (6Q)

9.15.1 Conditional Probability: GATE2013 AE: GA-10

In a factory, two machines $M1$ and $M2$ manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of $M1$ and 3% of $M2$ are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by $M2$?

A. 0.35 B. 0.45 C. 0.5 D. 0.4

gate2013-ae quantitative-aptitude conditional-probability

Answer key

9.15.2 Conditional Probability: GATE2014 AG: GA-10

10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is

_______.

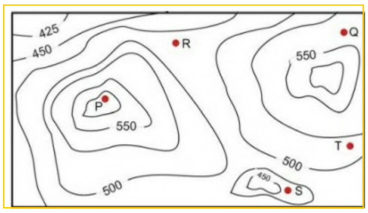
gate2014-ag quantitative-aptitude probability conditional-probability normal numerical-answers

Answer key

9.16 Contour Plots (2)

9.16.1 Contour Plots: GATE CSE 2017 | Set 1 | GA | Question: 10

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



A. P, Q B. P, Q, T C. R, S, T D. Q, R, S

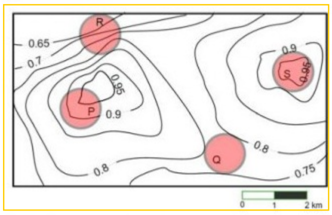
gatecse-2017-set1 general-aptitude quantitative-aptitude data-interpretation normal contour-plots

Answer key

9.16.2 Contour Plots: GATE CSE 2017 | Set 2 | GA | Question: 10

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?



A. P B. Q C. R D. S

gatecse-2017-set2 quantitative-aptitude data-interpretation normal contour-plots

Answer key

9.17 Cost Market Price (4)

✍ **Practice Test:** Test 1 (7Q)

9.17.1 Cost Market Price: GATE CSE 2011 | Question: 63

The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = \dfrac{100}{q}$. How many units should be produced to minimize the total cost $(V + F)$?

A. 5 B. 4 C. 7 D. 6

gatecse-2011 quantitative-aptitude cost-market-price normal

Answer key

9.17.2 Cost Market Price: GATE CSE 2012 | Question: 56

The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹50 per unit. The number of units to be produced by the firm such that the profit is maximized is

A. 5 B. 10 C. 15 D. 25

gatecse-2012 quantitative-aptitude cost-market-price normal

Answer key

9.17.3 Cost Market Price: GATE CSE 2019 | GA | Question: 4

Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs. 150 more. The cost of the gift was Rs. _____

A. 666 B. 3000 C. 6000 D. 12000

gatecse-2019 general-aptitude quantitative-aptitude cost-market-price one-mark

Answer key

9.17.4 Cost Market Price: GATE2014 AE: GA-5

A foundry has a fixed daily cost of $\text{Rs } 50,000$ whenever it operates and a variable cost of $\text{RS } 800Q$, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes.

gate2014-ae quantitative-aptitude cost-market-price numerical-answers

Answer key

9.18 Cubes (2)

9.18.1 Cubes: GATE CSE 2016 | Set 1 | GA | Question: 05

A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is __________.

a. 56 b. 64 c. 72 d. 96

gatecse-2016-set1 quantitative-aptitude geometry normal cubes

Answer key

9.18.2 Cubes: GATE CSE 2021 | Set 2 | GA | Question: 3

If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos\theta =$

A. $\frac{1}{2}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{1}{\sqrt{2}}$ D. $\frac{\sqrt{3}}{2}$

gatecse-2021-set2 quantitative-aptitude mensuration cubes one-mark

Answer key

9.19 Currency Notes (1)

9.19.1 Currency Notes: GATE2012 CY: GA-10

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

A. 5 B. 6 C. 9 D. 10

gate2012-cy quantitative-aptitude numerical-computation currency-notes

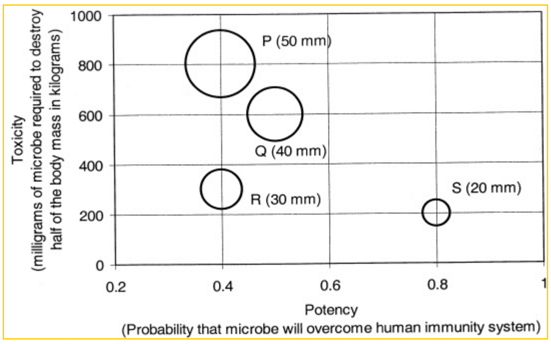
Answer key

9.20 Data Interpretation (3)

✍ **Practice Test:** Test 1 (8Q)

9.20.1 Data Interpretation: GATE CSE 2011 | Question: 62

P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

A. P B. Q C. R D. S

gatecse-2011 quantitative-aptitude data-interpretation normal

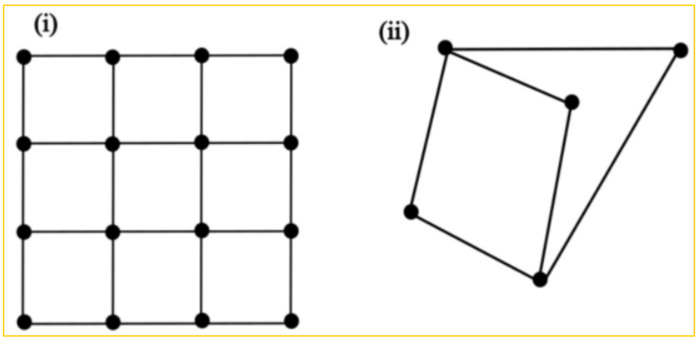
Answer key

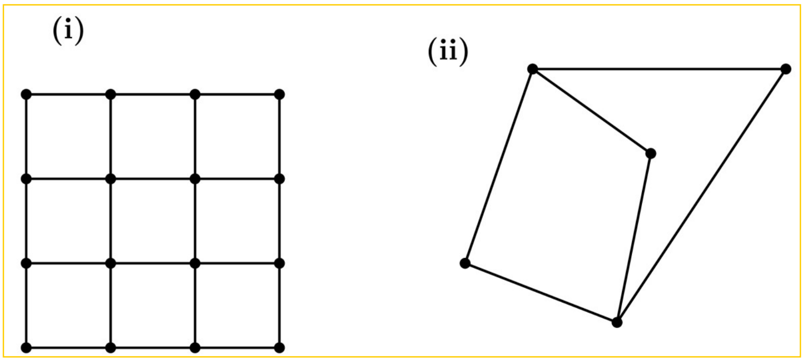
9.20.2 Data Interpretation: GATE CSE 2026 | Set 2 | GA | Question: 8

Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways.

A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started.

Which one of the following options is then true?





A. Such a trip is possible for (i), but not for (ii).
B. Such a trip is possible for (ii), but not for (i).
C. Such a trip is possible for both (i) and (ii).
D. Such a trip is possible neither for (i) nor for (ii).

gatecse-2026-set2 quantitative-aptitude data-interpretation two-marks

Answer key

9.20.3 Data Interpretation: GATE DA 2025 | GA | Question: 7

Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$.

Which one of the following statements is correct about their age on the day when they attain the same weight?

A. $a_1 < a_2 < 50$
B. $a_1 < 50 < a_2$
C. $50 < a_1 < a_2$
D. Either $a_1 = 50$ or $a_2 = 50$

gateda-2025 quantitative-aptitude data-interpretation two-marks

Answer key

9.21 Digital Image Processing (1)

9.21.1 Digital Image Processing: GATE DA 2025 | GA | Question: 3

A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

A. 3
B. 8
C. 11
D. 9

gateda-2025 quantitative-aptitude digital-image-processing easy one-mark

Answer key

9.22 Factors (3)

✍ **Practice Test:** Test 1 (5Q)

9.22.1 Factors: GATE CSE 2013 | Question: 62

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7 ?

A. $\left(\frac{13}{90}\right)$

B. $\left(\frac{12}{90}\right)$

C. $\left(\frac{78}{90}\right)$

D. $\left(\frac{77}{90}\right)$

gatecse-2013 quantitative-aptitude easy probability factors

Answer key

9.22.2 Factors: GATE CSE 2018 | GA | Question: 4

What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

A. 3047

B. 6047

C. 7987

D. 63847

gatecse-2018 quantitative-aptitude factors one-mark

Answer key

9.22.3 Factors: GATE2010 MN: GA-8

Consider the set of integers $\{1, 2, 3, \ldots, 5000\}$. The number of integers that is divisible by neither 3 nor 4 is :

A. 1668

B. 2084

C. 2500

D. 2916

general-aptitude quantitative-aptitude gate2010-mn factors

Answer key

9.23 Fractions (1)

9.23.1 Fractions: GATE CSE 2018 | GA | Question: 6

In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift $Rs\ 750$ to each male senior citizen in the town and $Rs\ 1000$ to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\frac{8}{9}^{\text{th}}$ of the eligible men and $\frac{2}{3}^{\text{rd}}$ of the eligible women claimed the gift.How much money (in Rupees) did the philanthropist give away in total?

A. $1,50,000$

B. $2,00,000$

C. $1,75,000$

D. $1,51,000$

gatecse-2018 quantitative-aptitude fractions normal two-marks

Answer key

9.24 Functions (7)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (7Q)

9.24.1 Functions: GATE CSE 2015 | Set 2 | GA | Question: 9

If p, q, r, s are distinct integers such that:

$$f(p,q,r,s) = \max(p,q,r,s)$$

$$g(p,q,r,s) = \min(p,q,r,s)$$

$h(p,q,r,s) =$ remainder of $\frac{(p \times q)}{(r \times s)}$ if $(p \times q) > (r \times s)$

or remainder of $\frac{(r \times s)}{(p \times q)}$ if $(r \times s) > (p \times q)$

Also a function $fgh(p,q,r,s) = f(p,q,r,s) \times g(p,q,r,s) \times h(p,q,r,s)$
Also the same operations are valid with two variable functions of the form $f(p,q)$
What is the value of $fg\left(h\left(2,5,7,3\right),4,6,8\right)$?

gatecse-2015-set2 functions normal numerical-answers

Answer key

9.24.2 Functions: GATE CSE 2015 | Set 3 | GA | Question: 5

A function $f(x)$ is linear and has a value of 29 at $x=-2$ and 39 at $x=3$. Find its value at $x=5$.

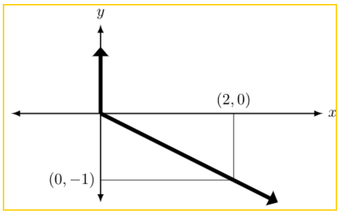
A. 59
B. 45
C. 43
D. 35

gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.24.3 Functions: GATE CSE 2015 | Set 3 | GA | Question: 8

Choose the most appropriate equation for the function drawn as thick line, in the plot below.



A. $x=y-|y|$
B. $x=-(y-|y|)$
C. $x=y+|y|$
D. $x=-(y+|y|)$

gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.24.4 Functions: GATE CSE 2023 | GA | Question: 7

Consider two functions of time (t),

$$f(t)=0.01t^2$$
$$g(t)=4t$$

where $0<t<\infty$.

Now consider the following two statements:

i. For some $t>0, g(t)>f(t)$.
ii. There exists a T, such that $f(t)>g(t)$ for all $t>T$.

Which one of the following options is TRUE?

A. only (i) is correct
B. only (ii) is correct
C. both (i) and (ii) are correct
D. neither (i) nor (ii) is correct

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.24.5 Functions: GATE CSE 2023 | GA | Question: 9

$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x)=g(y)$ for all values of x and y. Which one of the following options is necessarily TRUE for a x and y?

A. $f(x)=0$ and $g(y)=0$
B. $f(x)=g(y)=$ constant
C. $f(x)\neq$ constant and $g(y)\neq$ constant
D. $f(x)+g(y)=f(x)-g(y)$

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.24.6 Functions: GATE2010 MN: GA-10

Given the following four functions $f_1(n)=n^{100}$, $f_2(n)=(1.2)^n$, $f_3(n)=2^{n/2}$, $f_4(n)=3^{n/3}$ which function will have the largest value for sufficiently large values of n $(i.e.\ n\rightarrow\infty)$?

A. f_4 B. f_3 C. f_2 D. f_1

general-aptitude quantitative-aptitude gate2010-mn functions

Answer key

9.24.7 Functions: GATE2012 AR: GA-7

Let $f(x) = x - [x]$, where $x \geq 0$ and $[x]$ is the greatest integer not larger than x. Then $f(x)$ is a

A. monotonically increasing function
B. monotonically decreasing function
C. linearly increasing function between two integers
D. linearly decreasing function between two integers

gate2012-ar quantitative-aptitude functions normal

Answer key

9.25 Geometry (5)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (4Q)

9.25.1 Geometry: GATE CSE 2014 | Set 1 | GA | Question: 10

When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines?

gatecse-2014-set1 quantitative-aptitude geometry combinatory normal numerical-answers

Answer key

9.25.2 Geometry: GATE CSE 2024 | Set 1 | GA | Question: 7

A rectangular paper sheet of dimensions $54\text{ cm} \times 4\text{ cm}$ is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.

Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

A. $1/\pi$ B. $2/\pi$ C. $3/\pi$ D. $4/\pi$

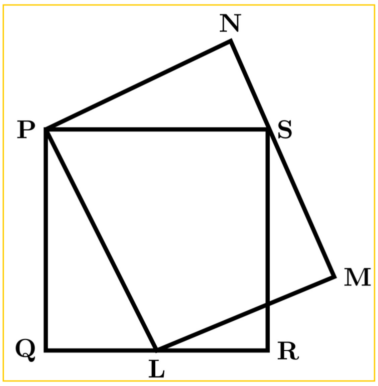
gatecse-2024-set1 quantitative-aptitude geometry two-marks

Answer key

9.25.3 Geometry: GATE CSE 2025 | Set 2 | GA | Question: 9

In the Given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square.

What is the area (in cm^2) of the rectangle PLMN?
Note: The figure shown is representative.



A. $2\sqrt{2}$

B. 2

C. 8

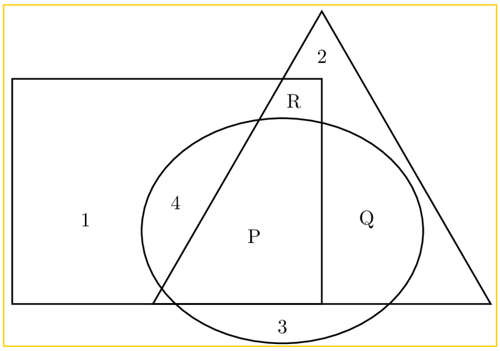
D. 4

gatecse2025-set2 quantitative-aptitude geometry two-marks

Answer key☟

9.25.4 Geometry: GATE DA 2025 | GA | Question: 4

In the given figure, the numbers associated with the rectangle, triangle, and ellipse are $1, 2$, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R ?



A. $P=6; Q=5; R=3$

B. $P=5; Q=6; R=3$

C. $P=3; Q=6; R=6$

D. $P=5; Q=3; R=6$

gateda-2025 quantitative-aptitude geometry one-mark

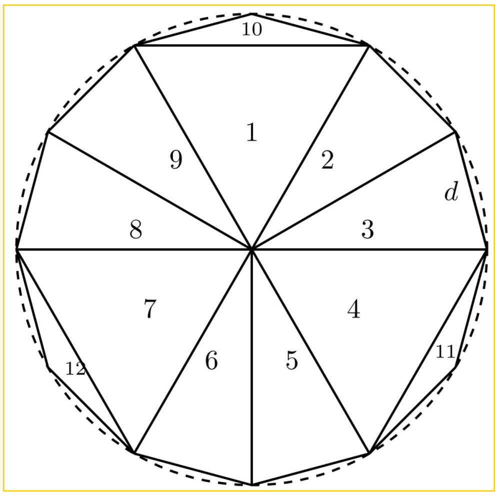
Answer key☟

9.25.5 Geometry: GATE DA 2025 | GA | Question: 8

A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square.

The number of squares that can be formed and the number of triangles required to form each square, respectively, are:

Note: The figure shown is representative.



A. $3; 4$

B. $4; 3$

C. $3; 3$

D. $3; 2$

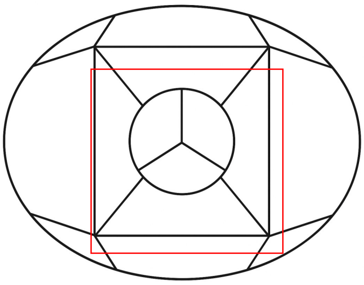
gateda-2025 quantitative-aptitude geometry two-marks

Answer key☟

9.26 Graph Coloring (1)

9.26.1 Graph Coloring: GATE DS&AI 2024 | GA | Question: 2

The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is



A. 4 B. 3 C. 5 D. 6

gate-ds-ai-2024 quantitative-aptitude graph-coloring one-mark easy

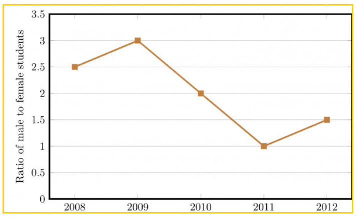
Answer key

9.27 Line Graph (5)

Practice Test: Test 1 (12Q)

9.27.1 Line Graph: GATE CSE 2014 | Set 2 | GA | Question: 9

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

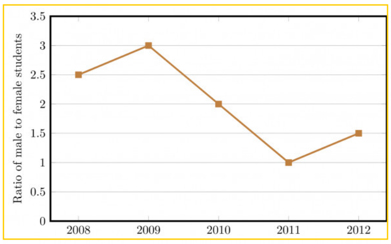


gatecse-2014-set2 quantitative-aptitude data-interpretation numerical-answers normal line-graph

Answer key

9.27.2 Line Graph: GATE CSE 2014 | Set 3 | GA | Question: 9

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?

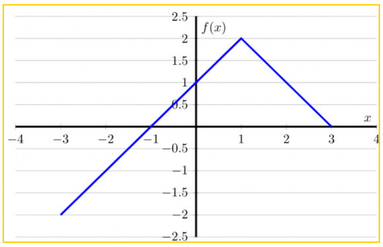


A. 1 : 1 B. 2 : 1 C. 1.5 : 1 D. 2.5 : 1

gatecse-2014-set3 quantitative-aptitude data-interpretation normal line-graph

Answer key

9.27.3 Line Graph: GATE CSE 2016 | Set 2 | GA | Question: 10



A. $f(x) = 1 - |x - 1|$

B. $f(x) = 1 + |x - 1|$

C. $f(x) = 2 - |x - 1|$

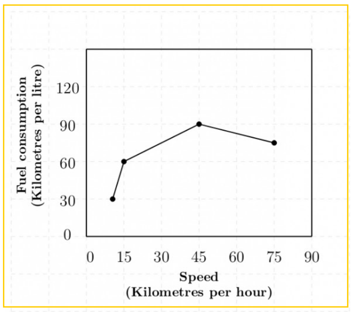
D. $f(x) = 2 + |x - 1|$

gatecse-2016-set2 quantitative-aptitude data-interpretation normal line-graph

Answer key

9.27.4 Line Graph: GATE2011 AG: GA-9

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average Speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

A. P

B. Q

C. R

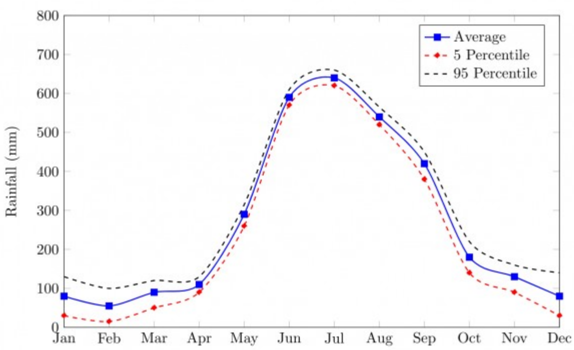
D. S

general-aptitude quantitative-aptitude gate2011-ag data-interpretation line-graph

Answer key

9.27.5 Line Graph: GATE2014 AE: GA-10

The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure.



Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)

i. On average, it rains more in July than in December
ii. Every year, the amount of rainfall in August is more than that in January
iii. July rainfall can be estimated with better confidence than February rainfall
iv. In August, there is at least 500 mm of rainfall

A. (i) and (ii)
B. (i) and (iii)
C. (ii) and (iii)
D. (iii) and (iv)

gate2014-ae quantitative-aptitude data-interpretation line-graph

Answer key

9.28 Logarithms (6)

Practice Tests: Test 1 (15Q) Test 2 (4Q)

9.28.1 Logarithms: GATE CSE 2011 | Question: 57

If $\log(\mathrm{P}) = (1/2)\log(\mathrm{Q}) = (1/3)\log(\mathrm{R})$, then which of the following options is **TRUE**?

A. $\mathrm{P}^2 = \mathrm{Q}^3\mathrm{R}^2$
B. $\mathrm{Q}^2 = \mathrm{PR}$
C. $\mathrm{Q}^2 = \mathrm{R}^3\mathrm{P}$
D. $\mathrm{R} = \mathrm{P}^2\mathrm{Q}^2$

gatecse-2011 quantitative-aptitude normal logarithms

Answer key

9.28.2 Logarithms: GATE CSE 2018 | GA | Question: 7

If $pqr \neq 0$ and $p^{-x} = \dfrac{1}{q}, q^{-y} = \dfrac{1}{r}, r^{-z} = \dfrac{1}{p}$, what is the value of the product xyz ?

A. -1
B. $\dfrac{1}{pqr}$
C. 1
D. pqr

gatecse-2018 quantitative-aptitude ratio-proportion two-marks logarithms

Answer key

9.28.3 Logarithms: GATE CSE 2024 | Set 1 | GA | Question: 5

For positive non-zero real variables p and q, if

$\log\left(p^2 + q^2\right) = \log p + \log q + 2\log 3,$

then, the value of $\frac{p^4+q^4}{p^2q^2}$ is

A. 79
B. 81
C. 9
D. 83

gatecse-2024-set1 quantitative-aptitude logarithms one-mark

Answer key

9.28.4 Logarithms: GATE CSE 2024 | Set 2 | GA | Question: 4

For positive non-zero real variables x and y, if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is

A. 1
B. $1/2$
C. 2
D. 4

gatecse-2024-set2 quantitative-aptitude logarithms one-mark

Answer key

9.28.5 Logarithms: GATE DA 2026 | GA | Question: 4

Consider two distinct positive real numbers m, n, with $m > n$. Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is ______.

A. $x > y$ B. $x < y$ C. $x = y$ D. $x = \log_{10}(y)$

gateda-2026 quantitative-aptitude logarithms one-mark

Answer key

9.28.6 Logarithms: GATE2012 AR: GA-6

A value of x that satisfies the equation $\log x + \log(x-7) = \log(x+11) + \log 2$ is

A. 1 B. 2 C. 7 D. 11

gate2012-ar quantitative-aptitude logarithms

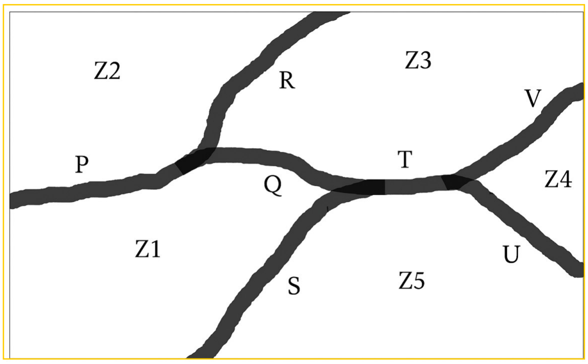
Answer key

9.29 Maps (1)

9.29.1 Maps: GATE CSE 2025 | Set 2 | GA | Question: 10

The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked $Z1, Z2, Z3, Z4$, and $Z5$. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct?

Note: The figure shown is representative.



A. Bridges on P, Q and T
B. Bridges on P, Q, S and T
C. Bridges on Q, R, T and V
D. Bridges on P, Q, S, U and V

gatecse2025-set2 quantitative-aptitude maps two-marks

Answer key

9.30 Maxima Minima (4)

✍ **Practice Test:** Test 1 (5Q)

9.30.1 Maxima Minima: GATE CSE 2012 | Question: 62

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters B. 10 meters C. 12 meters D. 14 meters

gatecse-2012 quantitative-aptitude normal maxima-minima

Answer key

9.30.2 Maxima Minima: GATE CSE 2017 | Set 2 | GA | Question: 9

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0
B. 1
C. 2
D. 3

gatecse-2017-set2 quantitative-aptitude normal maxima-minima calculus

Answer key

9.30.3 Maxima Minima: GATE2010 TF: GA-5

Consider the function $f(x) = \max(7 - x, x + 3)$. In which range does f take its minimum value?

A. $-6 < x < -2$
B. $-2 \leq x < 2$
C. $2 \leq x < 6$
D. $6 \leq x < 10$

general-aptitude quantitative-aptitude gate2010-tf maxima-minima functions

Answer key

9.30.4 Maxima Minima: GATE2013 CE: GA-6

X and Y are two positive real numbers such that $2X + Y \leq 6$ and $X + 2Y \leq 8$. For which of the following values of (X, Y) the function $f(X, Y) = 3X + 6Y$ will give maximum value ?

A. $\left(\frac{4}{3}, \frac{10}{3}\right)$
B. $\left(\frac{8}{3}, \frac{20}{3}\right)$
C. $\left(\frac{8}{3}, \frac{10}{3}\right)$
D. $\left(\frac{4}{3}, \frac{20}{3}\right)$

gate2013-ce quantitative-aptitude maxima-minima

Answer key

9.31 Mensuration (1)

9.31.1 Mensuration: GATE DA 2025 | GA | Question: 5

A rectangle has a length L and a width W, where $\mathrm{L} > \mathrm{W}$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

A. Perimeter increases by 10%.
B. Length of the diagonals increases by 10%.
C. Area increases by 10%.
D. The rectangle becomes a square.

gateda-2025 quantitative-aptitude mensuration one-mark

Answer key

9.32 Modular Arithmetic (1)

9.32.1 Modular Arithmetic: GATE2012 CY: GA-1

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$

A. 2.23
B. 4.33
C. 11.37
D. 27.64

gate2012-cy quantitative-aptitude modular-arithmetic

Answer key

9.33 Number Representation (1)

9.33.1 Number Representation: GATE DA 2026 | GA | Question: 2

The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _____

A. 12
B. 14
C. 16
D. 18

gateda-2026 quantitative-aptitude one-mark number-representation

Answer key

9.34 Number Series (5)

✍ **Practice Test:** Test 1 (10Q)

9.34.1 Number Series: GATE CSE 2013 | Question: 61

Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots\ldots\ldots\ldots+\frac{1}{\sqrt{80}+\sqrt{81}}$$

A. 7 B. 8 C. 9 D. 10

gatecse-2013 quantitative-aptitude normal number-series

Answer key

9.34.2 Number Series: GATE CSE 2014 | Set 2 | GA | Question: 5

The value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$ is

A. 3.464 B. 3.932 C. 4.000 D. 4.444

gatecse-2014-set2 quantitative-aptitude easy number-series

Answer key

9.34.3 Number Series: GATE CSE 2014 | Set 3 | GA | Question: 4

Which number does not belong in the series below?

$$2, 5, 10, 17, 26, 37, 50, 64$$

A. 17 B. 37 C. 64 D. 26

gatecse-2014-set3 quantitative-aptitude number-series easy

Answer key

9.34.4 Number Series: GATE2010 TF: GA-7

Consider the series $\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{8}+\frac{1}{9}-\frac{1}{16}+\frac{1}{32}+\frac{1}{27}-\frac{1}{64}+\ldots$ The sum of the infinite series above is:

A. ∞ B. $\frac{5}{6}$ C. $\frac{1}{2}$ D. 0

general-aptitude quantitative-aptitude gate2010-tf number-series

Answer key

9.34.5 Number Series: GATE2013 EE: GA-10

Find the sum to $'n'$ terms of the series $10+84+734+\ldots$

A. $\frac{9(9^n+1)}{10}+1$ B. $\frac{9(9^n-1)}{8}+1$ C. $\frac{9(9^n-1)}{8}+n$ D. $\frac{9(9^n-1)}{8}+n^2$

gate2013-ee quantitative-aptitude number-series

Answer key

9.35 Number System (4)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (2Q)

9.35.1 Number System: GATE CSE 2014 | Set 3 | GA | Question: 10

Consider the equation: $(7526)_8-(Y)_8=(4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

A. 1634 B. 1737 C. 3142 D. 3162

gatecse-2014-set3 quantitative-aptitude number-system normal digital-logic

Answer key

9.35.2 Number System: GATE CSE 2017 | Set 2 | GA | Question: 8

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

A. 90 digits B. 91 digits C. 92 digits D. 93 digits

gatecse-2017-set2 quantitative-aptitude number-system

Answer key

9.35.3 Number System: GATE DA 2026 | GA | Question: 9

P, Q, R, S, X, and Y are distinct single-digit whole numbers taking values from 0 to 9.
PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number.

It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number.

The value of Y is ____

A. 4 B. 5 C. 6 D. 7

gateda-2026 quantitative-aptitude two-marks number-system

Answer key

9.35.4 Number System: GATE2010 MN: GA-9

A positive integer m in base 10 when represented in base 2 has the representation p and in base 3 has the representation q. We get $p - q = 990$ where the subtraction is done in base 10. Which of the following is necessarily true?

A. $m > 14$ B. $9 \leq m \leq 13$ C. $6 \leq m \leq 8$ D. $m < 6$

general-aptitude quantitative-aptitude gate2010-mn number-system

Answer key

9.36 Number Theory (2)

Practice Test: Test 1 (7Q)

9.36.1 Number Theory: GATE CSE 2025 | Set 2 | GA | Question: 3

If $P\,e^x = Q\,e^{-x}$ for all real values of x, which one of the following statements is true?

A. $P = Q = 0$ B. $P = Q = 1$ C. $P = 1; Q = -1$ D. $\frac{P}{Q} = 0$

gatecse2025-set2 quantitative-aptitude mathematical-logic one-mark number-theory

Answer key

9.36.2 Number Theory: GATE2012 AE: GA-8

If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

A. sum of squares of two natural numbers
B. sum of cubes of two natural numbers
C. sum of square roots of two natural numbers
D. sum of cube roots of two natural numbers

gate2012-ae number-theory quantitative-aptitude

Answer key

9.37 Numerical Computation (9)

Practice Tests: Test 1 (15Q) Test 2 (3Q)

9.37.1 Numerical Computation: GATE CSE 2014 | Set 1 | GA | Question: 4

If $\left(z+\frac{1}{z}\right)^2 = 98$, compute $\left(z^2+\frac{1}{z^2}\right)$.

gatecse-2014-set1 quantitative-aptitude easy numerical-answers numerical-computation

Answer key

9.37.2 Numerical Computation: GATE CSE 2014 | Set 2 | GA | Question: 4

What is the average of all multiples of 10 from 2 to 198?

A. 90
B. 100
C. 110
D. 120

gatecse-2014-set2 quantitative-aptitude easy numerical-computation factors

Answer key

9.37.3 Numerical Computation: GATE CSE 2017 | Set 1 | GA | Question: 4

Find the smallest number y such that $y \times 162$ is a perfect cube.

A. 24
B. 27
C. 32
D. 36

gatecse-2017-set1 general-aptitude quantitative-aptitude easy numerical-computation

Answer key

9.37.4 Numerical Computation: GATE CSE 2017 | Set 2 | GA | Question: 4

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

A. 12
B. 15
C. 18
D. 19

gatecse-2017-set2 quantitative-aptitude numerical-computation

Answer key

9.37.5 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 3

Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?

A. 127
B. 64
C. 63
D. 32

gatecse-2026-set1 quantitative-aptitude numerical-computation easy one-mark

Answer key

9.37.6 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 4

A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization courses. The maximum credits of elective courses that a student can enroll for is _____

A. 10 B. 15 C. 20 D. 25

gatecse-2026-set1 quantitative-aptitude numerical-computation one-mark

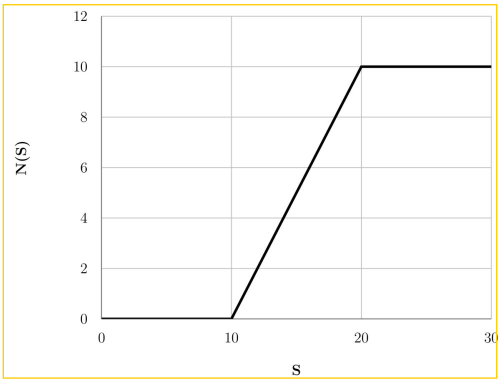
Answer key

9.37.7 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 8

For positive real numbers S and K, the function $H_K(S)$ is defined as:
$H_K(S) = \max(S - K, 0)$. The max function is defined as:

$$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$

The graph below shows the plot of a function $N(S)$ versus S.
$N(S)$ can be expressed as_____.



A. $H_{10}(S) - H_{20}(S)$
B. $H_{10}(S) - 2H_{20}(S)$
C. $-H_{10}(S) + H_{20}(S)$
D. $H_{15}(S) - H_{20}(S)$

gatecse-2026-set1 quantitative-aptitude numerical-computation two-marks

Answer key

9.37.8 Numerical Computation: GATE2010 TF: GA-10

A student is answering a multiple choice examination with 65 questions with a marking scheme as follows: $i)$ 1 marks for each correct answer, $ii)$ $-\frac{1}{4}$ for a wrong answer, $iii)$ $-\frac{1}{8}$ for a question that has not been attempted. If the student gets 37 marks in the test then the least possible number of questions the student has NOT answered is:

A. 6 B. 5 C. 7 D. 4

general-aptitude quantitative-aptitude gate2010-tf numerical-computation

Answer key

9.37.9 Numerical Computation: GATE2013 CE: GA-1

A number is as much greater than 75 as it is smaller than 117. The number is:

A. 91 B. 93 C. 89 D. 96

gate2013-ce quantitative-aptitude numerical-computation

Answer key

9.38 Percentage (8)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q)

9.38.1 Percentage: GATE CSE 2014 | Set 1 | GA | Question: 8

Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is $Rs\ 100$, a group of 5 tourists purchasing round-trip tickets will be charged Rs __________

gatecse-2014-set1 quantitative-aptitude easy numerical-answers percentage

Answer key☟

9.38.2 Percentage: GATE CSE 2014 | Set 3 | GA | Question: 8

The Gross Domestic Product (GDP) in Rupees grew at 7% during $2012-2013$. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period $2012-2013$ the exchange rate for the USD increased from $Rs.\ 50/USD$ to $Rs.60/USD$. India's GDP in USD during the period $2012-2013$

A. increased by 5%
B. decreased by 13%
C. decreased by 20%
D. decreased by 11%

gatecse-2014-set3 quantitative-aptitude normal percentage

Answer key☟

9.38.3 Percentage: GATE CSE 2024 | Set 1 | GA | Question: 4

The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio $5:3:13$. Of the total amount, the percentage of money in ₹5 coins is

A. 21%
B. $14\frac{2}{7}\%$
C. 10%
D. 30%

gatecse-2024-set1 quantitative-aptitude percentage one-mark

Answer key☟

9.38.4 Percentage: GATE2011 AG: GA-4

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

A. 100
B. 110
C. 90
D. 95

general-aptitude quantitative-aptitude gate2011-ag percentage

Answer key☟

9.38.5 Percentage: GATE2012 AE: GA-7

The total runs scored by four cricketers P, Q, R and S in years 2009 and 2010 are given in the following table;

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

A. P
B. Q
C. R
D. S

gate2012-ae quantitative-aptitude percentage

Answer key

9.38.6 Percentage: GATE2012 CY: GA-8

The data given in the following table summarizes the monthly budget of an average household.

Category	Amount (Rs.)
Food	4000
Rent	2000
Savings	1500
Other expenses	1800
Clothing	1200

The approximate percentage of the monthly budget **NOT** spent on savings is

A. 10% B. 14% C. 81% D. 86%

gate2012-cy quantitative-aptitude percentage

Answer key

9.38.7 Percentage: GATE2013 EE: GA-2

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was $41^\circ C$ and of Tuesday to Thursday was $43^\circ C$. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in $^\circ C$ on Thursday was

A. 40 B. 43 C. 46 D. 49

gate2013-ee quantitative-aptitude percentage

Answer key

9.38.8 Percentage: GATE2014 AE: GA-9

One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft?

A. 3.0 B. 2.5 C. 1.5 D. 1.25

gate2014-ae percentage quantitative-aptitude

Answer key

9.39 Permutation and Combination (4)

Practice Test: Test 1 (14Q)

9.39.1 Permutation and Combination: GATE CSE 2024 | Set 2 | GA | Question: 2

Two wizards try to create a spell using all the four elements, *water*, *air*, *fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.

How many attempts does each wizard make before coming to this conclusion, independently?

A. 24 B. 48 C. 16 D. 12

gatecse-2024-set2 quantitative-aptitude permutation-and-combination one-mark

Answer key

9.39.2 Permutation and Combination: GATE CSE 2025 | Set 1 | GA | Question: 10

A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

A. 4 B. 20 C. 24 D. 48

gatecse2025-set1 quantitative-aptitude permutation-and-combination two-marks

Answer key

9.39.3 Permutation and Combination: GATE DA 2026 | GA | Question: 7

Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20.

Ignoring permutations, the number of ways to pick these five integers is____

A. 0 B. 1 C. 2 D. 3

gateda-2026 quantitative-aptitude permutation-and-combination two-marks

Answer key

9.39.4 Permutation and Combination: GATE DS&AI 2024 | GA | Question: 3

How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1,3,4,6,7\}$, such that no digit appears more than once in a number?

A. 24 B. 48 C. 72 D. 12

gate-ds-ai-2024 quantitative-aptitude permutation-and-combination one-mark

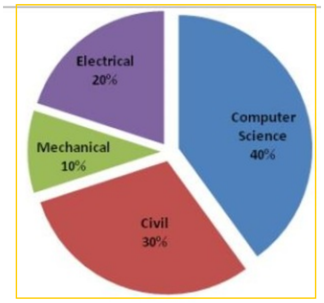
Answer key

9.40 Pie Chart (5)

Practice Tests: Test 1 (15Q) Test 2 (7Q)

9.40.1 Pie Chart: GATE CSE 2015 | Set 1 | GA | Question: 9

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is $5:4$. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the civil department and the female students in the Mechanical department?



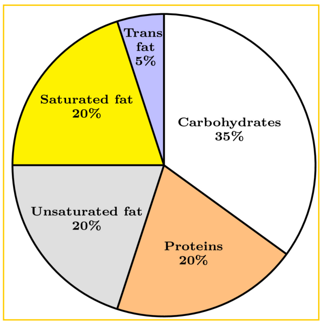
gatecse-2015-set1 quantitative-aptitude data-interpretation numerical-answers pie-chart

Answer key

9.40.2 Pie Chart: GATE CSE 2024 | Set 1 | GA | Question: 8

The pie chart presents the percentage contribution of different macronutrients to a typical $2,000$ kcal diet of a person.

Macronutrient energy contribution



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

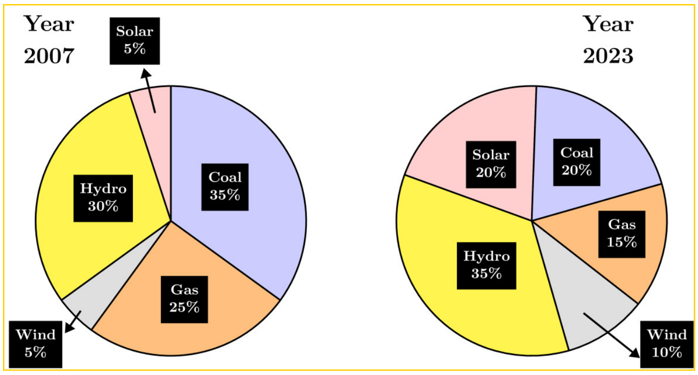
A. 44.4 B. 77.8 C. 100 D. $3,600$

gatecse-2024-set1 quantitative-aptitude pie-chart two-marks

Answer key

9.40.3 Pie Chart: GATE CSE 2024 | Set 2 | GA | Question: 8

The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

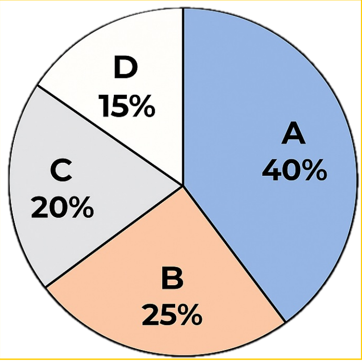
A. 25% B. 50% C. 77.5% D. 62.5%

gatecse-2024-set2 quantitative-aptitude pie-chart two-marks

Answer key

9.40.4 Pie Chart: GATE DS&AI 2024 | GA | Question: 5

In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were $1,15,000$, out of which $5,000$ were invalid.



Based on the data provided, the total number of valid votes received by the candidates B and C is

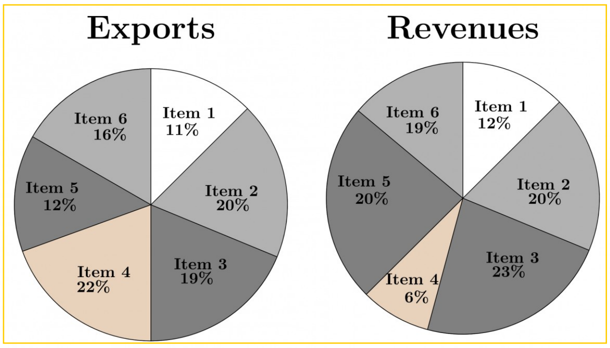
A. $45,000$ B. $49,500$ C. $51,750$ D. $54,000$

gate-ds-ai-2024 quantitative-aptitude pie-chart one-mark

Answer key

9.40.5 Pie Chart: GATE2014 AG: GA-8

The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?



A. $1:2$ B. $2:1$ C. $1:4$ D. $4:1$

gate2014-ag quantitative-aptitude data-interpretation pie-chart ratio-proportion normal

Answer key

9.41 Polynomials (1)

Practice Test: Test 1 (5Q)

9.41.1 Polynomials: GATE CSE 2016 Set 1 | GA | Question: 09

If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

A. $(x^3 + 8)$ B. $(x - 1)$ C. $(2x - 5)$ D. $(x + 1)$

gatecse-2016-set1 quantitative-aptitude polynomials normal

Answer key

9.42 Powers (1)

9.42.1 Powers: GATE DA 2025 | GA | Question: 9

If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:

A. 2^{-1} B. 2^0 C. 2^3 D. 2^{15}

gateda-2025 quantitative-aptitude powers two-marks

Answer key

9.43 Prime Numbers (1)

9.43.1 Prime Numbers: GATE CSE 2025 | Set 2 | GA | Question: 5

Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statemetns is correct for all values of p_1 and p_2?

A. $p_1 + p_2$ not a prime number
B. $p_1 p_2$ is not a prime number
C. $p_1 + p_2 + 1$ is a prime number
D. $p_1 p_2 + 1$ is a prime number

gatecse2025-set2 quantitative-aptitude easy prime-numbers one-mark

Answer key

9.44 Probability (17)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (10Q)

9.44.1 Probability: GATE CSE 2015 Set 1 | GA | Question: 10

The probabilities that a student passes in mathematics, physics and chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. $p + m + c = 27/20$
II. $p + m + c = 13/20$
III. $(p) \times (m) \times (c) = 1/10$

A. Only relation I is true.
B. Only relation II is true.
C. Relations II and III are true.
D. Relations I and III are true.

gatecse-2015-set1 quantitative-aptitude probability

Answer key

9.44.2 Probability: GATE CSE 2015 | Set 1 | GA | Question: 3

Given Set $A = 2, 3, 4, 5$ and Set $B = 11, 12, 13, 14, 15,$ two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

A. 0.20 B. 0.25 C. 0.30 D. 0.33

gatecse-2015-set1 quantitative-aptitude probability normal

Answer key

9.44.3 Probability: GATE CSE 2017 | Set 1 | GA | Question: 5

The probability that a k-digit number does NOT contain the digits $0, 5,$ or 9 is

A. 0.3^k B. 0.6^k C. 0.7^k D. 0.9^k

gatecse-2017-set1 general-aptitude quantitative-aptitude probability easy

Answer key

9.44.4 Probability: GATE CSE 2017 | Set 2 | GA | Question: 5

There are 3 red socks, 4 green socks and 3 blue socks.You choose 2 socks. The probability that they are of the same colour is

A. $\frac{1}{5}$
B. $\frac{7}{30}$
C. $\frac{1}{4}$
D. $\frac{4}{15}$

gatecse-2017-set2 quantitative-aptitude probability

Answer key

9.44.5 Probability: GATE CSE 2018 | GA | Question: 10

A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

A. Three green faces and four red faces.
B. Four green faces and three red faces.
C. Five green faces and two red faces.
D. Six green faces and one red face

gatecse-2018 quantitative-aptitude probability normal two-marks

Answer key

9.44.6 Probability: GATE CSE 2021 | Set 1 | GA | Question: 8

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is ___________

A. 0.3024
B. 0.4235
C. 0.6976
D. 0.8125

gatecse-2021-set1 quantitative-aptitude probability two-marks

Answer key

9.44.7 Probability: GATE CSE 2022 | GA | Question: 8

A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.

First ball is drawn. What is the probability of getting an orange ball in the next draw?

A. $\frac{1}{2}$
B. $\frac{8}{25}$
C. $\frac{19}{50}$
D. $\frac{23}{50}$

gatecse-2022 quantitative-aptitude probability two-marks

Answer key

9.44.8 Probability: GATE CSE 2025 | Set 1 | GA | Question: 8

A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

A. $\frac{75}{216}$
B. $\frac{1}{6}$
C. $\frac{1}{18}$
D. $\frac{25}{216}$

gatecse2025-set1 quantitative-aptitude probability easy two-marks

Answer key

9.44.9 Probability: GATE CSE 2026 | Set 1 | GA | Question: 10

An unbiased six-faced dice whose faces are marked with numbers $1, 2, 3, 4, 5,$ and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____

A. $\frac{1}{6}$
B. $\frac{5}{18}$
C. $\frac{7}{18}$
D. $\frac{5}{6}$

gatecse-2026-set1 quantitative-aptitude probability two-marks

Answer key

9.44.10 Probability: GATE CSE 2026 | Set 2 | GA | Question: 10

An unbiased six-faced dice whose faces are marked with numbers $1, 2, 3, 4, 5,$ and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is ________

A. $\frac{3}{36}$ B. $\frac{13}{36}$ C. $\frac{15}{36}$ D. $\frac{19}{36}$

gatecse-2026-set2 quantitative-aptitude probability two-marks

Answer key

9.44.11 Probability: GATE CSE 2026 | Set 2 | GA | Question: 3

A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

A. $1/4$ B. $3/4$ C. $2/3$ D. $3/8$

gatecse-2026-set2 quantitative-aptitude probability one-mark

Answer key

9.44.12 Probability: GATE DS&AI 2024 | GA | Question: 7

The probability of a boy or a girl being born is $1/2$. For a family having only three children, what is the probability of having two girls and one boy?

A. $3/8$ B. $1/8$ C. $1/4$ D. $1/2$

gate-ds-ai-2024 quantitative-aptitude probability two-marks

Answer key

9.44.13 Probability: GATE2012 AE: GA-6

Two policemen, A and B, fire once each at the same time at an escaping convict. The probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

A. 0.14 B. 0.22 C. 0.33 D. 0.40

gate2012-ae quantitative-aptitude probability

Answer key

9.44.14 Probability: GATE2012 AR: GA-9

A smuggler has 10 capsules in which five are filled with narcotic drugs and the rest contain the original medicine. All the 10 capsules are mixed in a single box, from which the customs officials picked two capsules at random and tested for the presence of narcotic drugs. The probability that the smuggler will be caught is

A. 0.50 B. 0.67 C. 0.78 D. 0.82

gate2012-ar quantitative-aptitude probability

Answer key

9.44.15 Probability: GATE2012 CY: GA-7

A and B are friends. They decide to meet between 1:00 pm and 2:00 pm on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

A. $1/4$ B. $1/16$ C. $7/16$ D. $9/16$

gate2012-cy quantitative-aptitude probability

Answer key

9.44.16 Probability: GATE2013 EE: GA-6

What is the chance that a leap year, selected at random, will contain 53 Saturdays?

A. $2/7$ B. $3/7$ C. $1/7$ D. $5/7$

gate2013-ee quantitative-aptitude probability

Answer key

9.44.17 Probability: GATE2014 AG: GA-4

In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is ____ years.

gate2014-ag quantitative-aptitude probability numerical-answers normal

Answer key

9.45 Profit Loss (3)

✍ Practice Test: Test 1 (14Q)

9.45.1 Profit Loss: GATE CSE 2021 | Set 1 | GA | Question: 7

Items	Cost (₹)	Profit \%	Marked Price (₹)
P	$5,400$	$---$	$5,860$
Q	$---$	25	$10,000$

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is $3:4$. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$(\text{Profit}\% = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100)$$

The discount on item Q, as a percentage of its marked price, is ________

A. 25 B. 12.5 C. 10 D. 5

gatecse-2021-set1 quantitative-aptitude profit-loss two-marks

Answer key

9.45.2 Profit Loss: GATE CSE 2024 | Set 2 | GA | Question: 7

A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

A. 1% profit B. 2% profit C. 1% loss D. 2% loss

gatecse-2024-set2 quantitative-aptitude profit-loss two-marks

Answer key

9.45.3 Profit Loss: GATE2013 CE: GA-9

A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

A. 5 B. 10 C. 15 D. 30

quantitative-aptitude gate2013-ce profit-loss

Answer key

9.46 Quadratic Equations (6)

Practice Test: Test 1 (13Q)

9.46.1 Quadratic Equations: GATE CSE 2014 | Set 1 | GA | Question: 5

The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b \mid x \mid +c = 0$ has

A. no roots B. 2 real roots C. 3 real roots D. 4 real roots

gatecse-2014-set1 quantitative-aptitude quadratic-equations normal

Answer key

9.46.2 Quadratic Equations: GATE CSE 2016 | Set 2 | GA | Question: 05

In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

A. n^4 B. 4^n C. 2^{2n-1} D. 4^{n-1}

gatecse-2016-set2 quantitative-aptitude quadratic-equations normal

Answer key

9.46.3 Quadratic Equations: GATE CSE 2021 | Set 2 | GA | Question: 4

If $\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = x + 2$, then the value of x is:

A. 2 B. 4 C. 6 D. 8

gatecse-2021-set2 quantitative-aptitude quadratic-equations one-mark

Answer key

9.46.4 Quadratic Equations: GATE CSE 2022 | GA | Question: 3

Let r be a root of the equation $x^2 + 2x + 6 = 0$.
Then the value of the expression $(r + 2)(r + 3)(r + 4)(r + 5)$ is

A. 51 B. -51 C. 126 D. -126

gatecse-2022 quantitative-aptitude quadratic-equations one-mark

Answer key

9.46.5 Quadratic Equations: GATE2011 MN: GA-62

A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as $(4, 3)$. In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as $(3, 2)$. Based on the above information, the roots of the correct quadratic equation are

A. $(-3, 4)$ B. $(3, -4)$ C. $(6, 1)$ D. $(4, 2)$

gate2011-mn quadratic-equations quantitative-aptitude

Answer key